

Post-Study Work Visa, Global Graduate Education and Skilled Immigrant Employment *

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Abstract

We study how tightening post-study work visa policy jointly shapes international students' decisions of where to study and where to work. We leverage the 2012 UK visa reform, which abolished a two-year unconditional post-study work visa with the aim of curbing low-skilled stay, to quantify how sensitive Indian master's students' education choices are to labor-market access in the destination country. Using a novel longitudinal panel of LinkedIn profiles, we first find that the policy brought large unintended hiring frictions in the high-skilled sector. Applying a difference-in-differences design, we further identify a significant decline in UK master's enrollment among Indian students. To interpret this response and separate its channels, we then develop a two-stage dynamic model of education and post-graduation work choices. The model attributes over one-third of the loss in high-skilled labor to the indirect deterrence effect on the education margin, a share that is even larger for STEM majors. We further find that a one-percent loss in skilled labor positions is associated with a 0.7% loss of students for STEM and a 0.5% loss for non-STEM. Our results highlight that a reform targeting low-skilled employment can create negative spillovers to high-skilled hiring, and quantify the cost of shrinking employment opportunities for education exports.

Keywords: Immigration Policy, Education, hiring friction

JEL Codes: F22, J61, I23

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1 Introduction

Between 2013 and 2019, global enrollment in international master’s programs increased from 1.2 to 1.7 million, with approximately 67 percent of international students originating from developing nations (OECD, *Education at a Glance 2021*). For these students, a foreign master’s degree serves as both a human capital investment and a strategic pathway to high-wage labor markets and permanent migration (Bound et al., 2021; Khanna, 2025). For host countries like the United States and the United Kingdom, international graduate education is a major export industry that cross-subsidizes domestic institutions (Shih, 2017; Bound et al., 2020; Zhu, 2024) and serves as a primary pipeline for high-skilled labor (Bound et al., 2015; Ruiz and Budiman, 2018). However, policymakers face competing political pressures to restrict overall immigration and prevent the potential displacement of native workers in lower-skilled occupations. To balance these trade-offs between retaining tuition revenue and global talent on the one hand, and controlling labor-market entry on the other, governments have implemented post-study work visa policies such as the OPT and H-1B regime in the United States.

The United Kingdom’s 2012 abolition of the Post-Study Work (PSW) visa represents a particularly sharp and highly consequential policy shock. Until 2012, international graduates of any UK higher-education institution were eligible for the Tier 1 PSW visa, which granted two years of unconditional residence with full labor-market access. However, the 2012 reform eliminated this route. Fresh graduates were only given a four-month grace period to secure a graduate-level (RQF-6) job offer from a government-certified sponsoring employer paying above a salary threshold, or to leave the country. While curbing visa abuse in low-skilled sectors is the major objective for the reform, the government also explicitly committed to continue attracting international students and retaining the “best and brightest” in skilled employment.

In this paper, we utilize novel data from LinkedIn profiles to understand the effect and mechanism of this policy reform. Specifically, we construct a longitudinal dataset of 94,606 Indian students, one of the largest sources of international master’s students, who pursued international master’s degrees between 2008 and 2013, linking their bachelor’s education, master’s country locations, and post-graduation outcomes. The descriptive analysis shows a sharp drop in low-skilled employment of Indian masters after the policy, but also a sharp decline in master enrollment and skilled employment. Exploiting state-level exposure variation of the policy measured by the pre-policy UK-stay rates, our Diff-in-Diff estimates reveal a 53% drop in U.K. master’s enrollments, translating to an estimated £110 million annual loss in tuition revenue only from Indian students. The heterogeneity analysis shows that

the decline in master enrollment is stronger for STEM students compared to non-STEM students.

To decompose the mechanisms driving the master’s enrollment and labor supply decline, we develop a two-stage discrete choice model that jointly captures students’ choice of master’s destination and their post-graduation labor market placement. In particular, we explicitly model labor market frictions by specifying job choice sets as stochastic across individuals. We establish the constructive identification of unobserved stochastic offer arrival probabilities from the joint policy response of master’s enrollment and post-graduation work decisions.¹

Using the model estimates, we find that the policy results in a 75.6% loss of high-skilled labor from Indian master’s students. The decomposition exercise shows that 63% of the total reduction in UK graduate-level employment is driven by the direct effect of increased search frictions on graduate-level UK employment. An additional 34 % is due to a forward-looking deterrence effect on the education enrollment margin: fewer students enroll in UK master’s programs because of their anticipation on higher post-graduation search frictions for graduate-level jobs. The remaining 3% reflects the enrollment deterrence caused by the complete removal of low-skilled employment options. Overall, Although the 2011 reform was advertised as targeting low-skilled stay, the shortened search window, sponsorship requirement, salary threshold, and administrative costs of certification jointly raised frictions on graduate-level hiring as well.

Consistent with our reduced-form evidence, the model estimates also show a larger education-enrollment response for STEM students than for non-STEM students. This gap is not due to a larger direct effect for the STEM group since it is even slightly larger for the Non-STEM (45%) compared to the STEM (40%). Instead, the elasticity of education enrollment to labor-market opportunity is much higher for STEM students than for non-STEM students, largely because STEM students face higher labor-market opportunity at baseline. Under a major-blind policy that imposes approximately the same proportional reduction in labor-market opportunity, the option value of a UK master’s therefore declines more for STEM students, making them the more sensitive group.

Finally, we quantify the price of shrinking employment opportunities for the education enrollment of Indian students, allowing us to predict the effects of reforms of differing magnitudes. We find that a one-percent loss of skilled labor positions reduces enrollment by

¹In a companion paper (Ma, 2026), we demonstrate that a misspecified full-choice-set model in which the friction enters as a utility rescaling rather than as a choice-set restriction, can introduce a bias into the predicted upstream response in either direction, depending on the magnitude of pre- and post-reform policy frictions and latent valuations. In our empirical setting, a misspecified full-choice-set model predicts an enrollment decline of only 22.4%, substantially understating the 41.9% contraction generated by our baseline specification with the friction channel.

0.7% for STEM students and 0.5% for non-STEM students. We also show that policymakers cannot predict this elasticity from the observed stay rate alone. Two families of primitive parameters that produce the same cross-sectional master’s-market share and stay rate can imply very different elasticities. The observed stay rate is a product of both availability and preference, and different weightings of the two, though generating the same observed stay rate, carry very different implications for students’ responsiveness to employment opportunity.

Our findings carry two broader implications for the design and evaluation of post-study immigration policy. First, eliminating visa pathways associated with lower-skilled employment, such as unconditional post-study residency periods, can substantially raise search and hiring frictions within the high-skilled sector. These flexible visa types provide valuable post-graduation job-search window, enabling highly qualified graduates to look for employer matches without the immediate risk of repatriation. In addition, these transitional pathways lower employer-side compliance costs, legal expenditures, and administrative risks during hiring phases. As a result, immigration policies designed to restrict lower-skilled labor pathways can generate unintended and strong frictions for high-skilled human capital retention.

Second, our findings emphasize the high cost of shrinking labor-market access for education enrollment, and hence for education-export revenue. This cost depends largely on baseline job availability and on a group’s share of the international education market, and it may therefore vary substantially across host countries and across student groups defined by major, ability, and source country. Predicting how any given group will respond requires separating **constraint** from **preference** in driving students’ post-graduation outcomes, since the same observed behavior can reflect very different underlying weights, and hence very different education responses to immigration policy.

This paper connects to four strands of literature. First, we contribute to the literature examining how immigration policy shapes international student enrollment. While it is well-established that open immigration policies attract international students while restrictive reforms deter them (Shih, 2016; Amuedo-Dorantes et al., 2019; Demirci, 2019; Amuedo-Dorantes et al., 2023; Chen et al., 2023; Abarcar and Theoharides, 2024; Hu and Winters, 2025)², previous studies primarily focus on the impact of visa policies on international enrollment and their subsequent downstream effects, such as generating college revenue (Bound et al., 2021; Zhu, 2024), stimulating local employment (Jia et al., 2025), and fostering regional entrepreneurship and innovation (Hunt, 2011; Stuen et al., 2012; Crown et al., 2020;

²A related literature examines other determinants of international education choice, including destination wages, migrant networks, human-capital portability, education quality, tuition and financial support, and completion rates (Rosenzweig et al., 2006; Beine et al., 2014, 2020; Arenas, 2021; Batista et al., 2025). Our structural model incorporates most of these determinants. We discuss this further in Section 5.

Beine et al., 2024; Ganguli and MacGarvie, 2025). For example, Zhu (2024) uses the same 2012 U.K. policy shock to demonstrate how restrictive U.K. visa rules diverted former British colonies’ international students to the United States, and subsequently increased domestic U.S. degree completion through tuition cross-subsidization. We complement this literature by explicitly tracing the channel through which post-study immigration policy first alters the hiring frictions and labor market access, and then shapes international education choices. Capturing these interconnected dynamics with a model allows us to quantify the magnitude of shrinking labor market access to education enrollment and education export revenue. Perhaps closest to our paper is Clemens et al. (2025), who similarly examine how immigration policy deters international students and subsequently shrinks the skilled workforce in the context of the U.S. However, our approach differs from theirs in two key dimensions. First, whereas they use survey methods to elicit the self-reported response of hypothetical work visa restrictions (e.g. OPT rescission), we employ a quasi-experimental approach to evaluate a realized but similar immigration reform in the U.K. Second, Clemens et al. (2025) impute the resulting labor supply shock by applying constant historical stay rates to a declining student pool. However, because stay rates themselves are endogenous to immigration policies, such an approach may blur the true labor market impact. Instead, we introduce a structural framework that jointly models the education enrollment and labor supply decision, capturing how policy reforms concurrently shift both the inflow of students and their post-graduation retention.³ Furthermore, our heterogeneity results also speak to papers on how policies alter the composition of the student body by field and ability (Kato and Sparber, 2013; Chen et al., 2023; Amuedo-Dorantes et al., 2023; Flynn et al., 2024).⁴ Documenting these heterogeneous responses matters because the success of immigration policies aimed at attracting and retaining talent depends on how the reform reshapes the composition of the international student body, not only its overall size. Our approach allows us to investigate the mechanisms driving heterogeneous impacts across student groups and to predict heterogeneous responses to other potential policies.

Second, we contribute to the literature concerning immigration policy and the skilled labor supply. Past studies have extensively investigated how visa policies affect the stock of skilled workers and subsequently impact native-born employment and wages (Peri et al.,

³The dynamic model in our paper is closest to Khanna and Morales (2025), in which Indian students make forward-looking decisions regarding their college majors, explicitly factoring in the probability of migrating to the U.S. to work as computer scientists.

⁴Amuedo-Dorantes et al. (2023) find that, following the OPT STEM extension, international matriculation in STEM bachelor’s and master’s programs increased by 18% and 30%, respectively, relative to non-STEM programs. They observe that these effects are most pronounced at highly selective institutions, though they note this may partially reflect demand-side responses (e.g., capacity expansion) rather than purely compositional changes in student choice.

2015a,b; Kerr et al., 2015; Bound et al., 2015; Demirci, 2019, 2020), firm behavior and performance (Dimmock et al., 2022; Glennon, 2024), host country innovation (Kerr and Lincoln, 2010; Bernstein et al., 2022), as well as their spillover effects on third countries (Brinatti and Guo, 2025).⁵ Yet the margin at which immigration policy affects skilled labor supply remains understudied, particularly for workers who first enter on a student visa. These study-to-work migrants are a primary and rapidly growing source of skilled labor in the United States (Bound et al., 2015; Ruiz and Budiman, 2018; Beine et al., 2023), and they exhibit exceptional innovative performance upon entering the labor market (Hunt, 2011).⁶ Restrictive immigration policy can divert these migrants at two distinct margins: the earlier education margin (the indirect, or deterrence, effect) and the later work-location margin upon graduation (the direct effect). Distinguishing between them is essential for policy design, as a dominant direct effect calls for labor-market remedies such as easing sponsorship and search frictions, whereas a dominant deterrence effect calls for education-side instruments such as tuition subsidies. In this paper, we explicitly quantify these two margins and find that the direct effect accounts for approximately two-thirds of the decline in graduate-level employment, while the indirect deterrence effect accounts for approximately one-third. In addition, our paper distinguishes between students’ transitions into skilled versus unskilled labor supply, recognizing that these two outcomes carry vastly different policy implications.⁷

Third, we contribute to a literature on graduate education that remains comparatively underdeveloped given the recent rapid global expansion of advanced degrees (Altonji et al., 2016). Most existing work studies the returns to graduate degrees for domestic students (Altonji and Zhong, 2021; Altonji and Zhu, 2025) while the returns for international master’s, which are potentially much larger due to access to higher-wage host-country labor markets, remain less examined. Chen (2024) is a notable exception, showing that Chinese firms place a significant premium on US credentials, which suggests that the returns to a foreign degree do not depend solely on staying abroad. While our primary focus is not on directly estimating these returns, our framework implicitly accounts for heterogeneous returns to

⁵A significant strand of literature also examines global talent flows and high-skilled immigrants’ location choices—most notably Amanzadeh et al. (2024), who also utilize LinkedIn data. These studies typically focus on the trade-offs between staying in a host country versus returning home, with broad implications for brain drain and brain gain (e.g., Adda et al., 2022). Yet, these studies typically remain silent on the preceding educational choices, concentrating instead on the final stage of the migration process. We depart from this approach by investigating the decision to acquire a master’s degree, incorporating future migration opportunities as a key determinant of global education choices.

⁶Bound et al. (2015) report that in 2010, 27% of all foreign-born workers and 35% of foreign-born IT workers in the United States initially arrived on a student visa, compared to 24% and 30% in 2003.

⁷Some research highlights how sectoral exemptions in visa policies shape immigrants’ career trajectories. For instance, Amuedo-Dorantes and Furtado (2019) demonstrates that because academic institutions are exempt from U.S. H-1B visa caps, strict private-sector visa constraints can push highly skilled foreign-born graduates to “settle” for academic jobs simply to remain in the developed country.

a master’s degree across different destination countries by linking expected wages to post-graduation work locations. Beyond the returns to education, recent work has explored the structural drivers of international educational mobility in the source country, showing how China’s economic development and domestic higher education expansion have fueled the surge in graduate enrollment at U.S. universities (Khanna et al., 2025; Jia et al., 2025). We complement this emerging literature by modeling students’ explicit choices of where to pursue their graduate education among competing global destinations, rather than treating enrollment in a single destination as the outcome of interest.

Finally, from a methodology angle, we contribute to the structural econometrics literature, in particular identification with a stochastic choice set. Observed choice probabilities alone cannot separately identify the arrival (or consideration) probability from the preference parameters governing acceptance. The literature typically restores identification through “identification at infinity” (Heckman, 1990) using an excluded shifter that drives the arrival or acceptance probability to the limits (French and Taber, 2011; Arcidiacono et al., 2025), or imposing some structure on choice decision to achieve partial identification (Manzini and Mariotti, 2014; Barseghyan et al., 2021).⁸ We propose a new point identification strategy that pairs an excluded policy shock with its dynamic upstream response. Because the shock shifts the arrival probability but leaves the utility of the option unaffected, the ratio of the responses across the two stages nets out the arrival margin and isolates the preference margin, achieving the identification of the latent value of the option. This identification framework applies broadly to other sequential-choice settings with a credibly excluded shifter of the downstream choice arrival probability, such as evaluating how tighter antitrust enforcement alters successful exit probabilities and subsequently deters upstream entrepreneurial entry.

The rest of the paper is organized as follows. Section 2 describes the institutional context of the UK master’s system and the work-visa routes. Section 3 introduces the LinkedIn profile data and provides descriptive facts about Indian master’s students. Section 4 presents the quasi-experimental evidence. Section 5 presents the model, and Section 6 describes identification and estimation. Section 7 reports the parameter estimates and model fit. Section 8 presents results from the counterfactual simulations, and Section 9 concludes.

⁸Similarly, in the job search model, the offer arrival rate is confounded with the flow value of nonemployment on the unemployment side (Flinn and Heckman, 1982), and with switching costs on the on-the-job-search side; the literature separates them using moments from the observed wage and acceptance-wage distributions or from firms’ equilibrium wage-setting response (Bontemps et al., 2000).

2 Institutional Context

This section details the institutional background of international graduate education in the United Kingdom around 2010. We first describe the UK master’s market for international students and the position of Indian students within it. Then, we outline the pre-2012 immigration regime and the subsequent policy reform that structurally altered the school-to-work transition.

2.1 UK Master Study and International students

UK is the second most popular host country for international students. In 2009, the UK hosted 10% of the world’s international students, second only to the United States, which hosted 18%.⁹ Within UK international student enrollment, “taught master’s” programs are the most common educational investment, accounting for 63.6% of international enrollments each year, compared to 21.0% for undergraduate programs.¹⁰ Taught master’s programs in the UK are typically one year in length, compared with the two-year structure that prevails in most US programs. Because the academic program was brief and intensive, students had limited capacity to engage in job searching, networking, or internships during their studies. Under the pre-2012 regime, this limitation was non-binding, as the two-year Post-Study Work (PSW) visa provided an ample post-graduation search horizon. However, this one-year academic structure would later interact disastrously with the 2012 policy’s four-month search window, imposing severe time constraints on job matching.

Entry into the U.K. education system was largely frictionless. International students admitted to a certified institution were, conditional on proof of funds and English-language competence, virtually guaranteed a Tier 4 (General) student visa. Instead, the main constraint is the tuition cost. Taught master’s programs charged international students average tuition of approximately £10,431 per year for the 2010 cohort, with substantial variation across institutions and fields. Because U.K. universities offer virtually no financial aid or

⁹Source: OECD (2011), *Education at a Glance 2011: OECD Indicators*, Chart C3.2.

¹⁰Enrollment shares are estimated from the Higher Education Statistics Agency (HESA) Student Record, Table 8 of the 2010/11 release, which reports the stock of non-UK domiciled students enrolled at UK higher education institutions by country of domicile and level of study. Because Table 8 reports the standard registration population (all currently enrolled students) rather than annual entrants, raw shares overstate the prevalence of multi-year programs. To approximate annual entrant shares, I divide the enrolled stock in each level of study by typical program length: 1 year for taught master’s degrees and other postgraduate programs, 3.5 years for research degrees (PhDs) and first degrees (averaging across 3-year programs in England/Wales/Northern Ireland and 4-year programs in Scotland), and 2 years for other undergraduate qualifications (e.g., HND, HNC, foundation degrees). The implied entrant shares are not highly sensitive to these assumptions: varying the first-degree length between 3 and 4 years shifts the taught master’s share by less than 2 percentage points.

tuition subsidies to international master’s students, these costs are paid almost entirely out-of-pocket or debt-financed by students and their families. Consequently, tuition represents a highly salient component of the cost-benefit calculation when selecting a destination. For these debt-financed students, the option value of accessing the high-wage U.K. labor market post-graduation serves as the fundamental economic justification for their educational investment.

Indian students, who constitute the primary sample of our analysis, represent the largest and most strategically responsive non-EU demographic in this market. In 2010, 39,090 Indian students studied in all levels of U.K. higher education institutions, making India the second-largest source country for U.K. international education export behind China (67,325). Indian enrollment was overwhelmingly concentrated at the graduate level: 87.5 percent of estimated annual entrants enrolled in taught master’s programs.¹¹ Furthermore, these students were disproportionately enrolled in STEM disciplines (XX percent). This heavy financial exposure and concentration in career-oriented STEM fields explains why Indian master’s students were exceptionally sensitive to downstream labor market frictions, making them an ideal population for identifying the joint elasticity of education and migration choices.

2.2 The Pre-2012 Regime: Frictionless Labor Market Entry

Prior to the 2012 reform, the U.K. operated a highly permissive immigration regime that offered multiple pathways for international graduates to enter the labor market. Although there are some negligible share of alternative pathways such as entrepreneurship visas, the three primary routes were the Tier 1 General, Tier 1 Post-Study Work (PSW) visas, and Tier 2 (General).

2.2.1 Tier 1 General

The Tier 1 (General) visa was a points-based route open to skilled applicants without an employer sponsor; eligibility was determined by age, education, and prior earnings. Although fresh graduates could in principle apply, the prior-earnings requirement made the route impractical for most international students entering the labor market for the first time. Figure 1, shows that the number of students transitioning from a study visa to a Tier 1 (General) visa was negligible in 2011.

¹¹This share is calculated using the same method and data source as for the overall international enrollment shares reported above.

2.2.2 Tier 1 PSW

The Tier 1 Post-Study Work (PSW) visa was the dominant pathway from international study to UK work. The PSW visa granted recent graduates of U.K. universities an unconditional right to live and work in the country for two years. During this window, graduates enjoyed unrestricted access to the labor market without requiring a firm to sponsor their visa. Because it circumvented employer sponsorship frictions entirely, students and employers overwhelmingly preferred the PSW route. As illustrated in Figure 1, prior to the reform, XX percent of students transitioning from a Tier 4 study visa to a work visa utilized the PSW route, compared to only XX percent transitioning directly to a Tier 2 visa.

2.2.3 Tier 2 Visa

The Tier 2 (General) visa coexisted with PSW and was the standard employer-sponsored route, like H-1B visa in the U.S. Prior to the reform, Tier 2 required an employer-sponsored job offer, an occupational skill level of NQF Level 6 or above, and a salary above a published threshold. The National Qualifications Framework (NQF) classified the educational qualifications of UK occupations by level of difficulty until 2015. The framework comprised eight main levels: levels 1–3 correspond roughly to secondary education, levels 4–6 to undergraduate study, level 7 to master’s degrees, and level 8 to doctoral degrees. In 2012, the minimum skill requirement for a Tier 2 visa was NQF level 6, the equivalent of a bachelor’s degree. Despite Tier 2’s availability pre-reform, the large majority of international graduates transitioned to PSW rather than directly to Tier 2 during 2008–2011. As shown in Figure 1, among all Tier 4 student visa holders transitioning to a work visa in, approximately [XX]% transitioned to Tier 1 (PSW) and only [XX]% transitioned to Tier 2 directly. Several institutional features explain this preference: PSW required no employer sponsor and no salary floor, allowed self-employment, and permitted graduates to search for employment without time pressure. By contrast, Tier 2 imposed sponsorship requirements that limited graduates to employers holding a Home Office license, and applied salary and skill thresholds that bound for some graduate-level occupations. For both students and employers, PSW was the lower-friction option.

2.2.4 Other Visa Routes

A small number of graduates entered the UK labor market through other Tier 1 sub-routes, most notably Tier 1 (Entrepreneur) and Tier 1 (Investor). These routes accounted for a quantitatively negligible share of post-study transitions during our sample period and we do not consider them further.

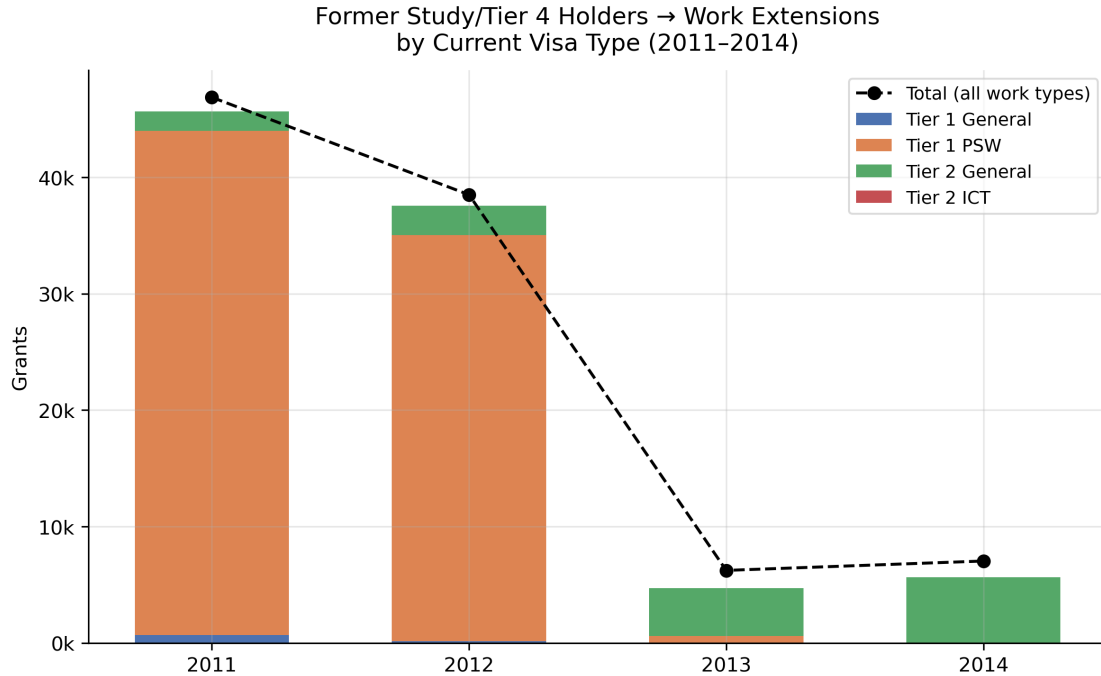


Figure 1: UK Visa Issuance

2.3 The 2011/2012 Reform: From PSW to tighter Tier 2

In early 2011, the U.K. government announced a comprehensive overhaul of its student immigration regime. The explicit objective of the reform was a dual mandate: to curb the perceived abuse of student visas in low-skilled sectors while selectively retaining top global talent. Justifying the closure of the unrestricted two-year search window, Theresa May, who was the Home Secretary at that time, argued that the previous system allowed “too many foreign graduates [to stay] on to work in unskilled jobs”¹². Crucially, however, policy-makers maintained a stated commitment to high-skilled retention; official policy directives emphasized a desire to attract the “best and the brightest,” explicitly aiming to design a system where “only the best of them” successfully transition into the domestic workforce upon graduation/¹³

The Tier 1 General Visa was officially closed to new applicants in April 2011. The abolition of the Tier 1 PSW visa was announced in March 2011 and fully implemented in April 2012. This timeline created a sharp exposure cut-off for international cohorts. Students who entered their one-year master’s programs in the fall of 2010 were largely unaffected, as they graduated in time to secure the PSW visa before its closure. Conversely, students enrolling in the fall of 2011 entered with full knowledge of that PSW would be unavailable upon their

¹²website here

¹³Website here

graduation. That motivates us to treat 2011 as the first treated cohort in our event-study analysis.

Upon the closure of the PSW route, fresh graduates were forced to rely exclusively on the Tier 2 visa, facing severe new time constraints. Depending on the length of their academic program, graduates were granted a grace period of only two months (for programs lasting 6 to 12 months) or four months (for programs lasting 12 months or more) to secure a sponsored job offer before being required to leave the country.

Beyond the compressed window, Tier 2 itself became substantially more restrictive, particularly due to two changes in Tier 2 Visa eligibility. First, the occupational skill threshold rose from NQF Level 3 to RQF Level 6, meaning that only graduate-level (bachelor’s-equivalent) occupations qualified. This change excluded a wide range of jobs that had previously been Tier-2-eligible. Second, the salary threshold tightened, with the Tier 2 (General) floor set at £20,800 — binding for some graduate-level occupations, particularly in non-STEM fields and in non-London labor markets. Sponsorship became more costly for employers as well. Reports from the House of Commons Home Affairs Committee indicate that during the first three quarters of 2012 alone, 469 Tier 2 sponsoring employers were suspended and 252 had their licenses permanently revoked, out of approximately 22,000 licensed sponsors — a revocation rate of roughly 3.3%, compared with approximately 2% during the first two years of the Tier 2 system (2008–2010). This increase reflected an expansion in post-license compliance visits, which reached 4,901 in the first three quarters of 2012, of which approximately 40% were unannounced — compared with only approximately 15% on the new applicants before 2011. These compliance costs deterred sponsorship at the margin, particularly among smaller and less specialized employers.

Overall, the 2012 reform replaced a frictionless two-year search option with a highly constrained, heavily audited employer-sponsorship regime. As shown in Figure 1, the number of visas issued for the transition from a Tier 4 study visa to a work visa dropped sharply following the implementation of the policy. While there was a marginal increase in Tier 2 General visa issuances, this substitution was entirely insufficient to compensate for the massive loss of the Tier 1 PSW visa, resulting in a severe aggregate decline in the retention of international graduates.

3 Data and Descriptive Statistics

3.1 LinkedIn Profile Data

To jointly understand the policy’s impact on education and work choice, we need to use longitudinal microdata tracking individuals across international borders. Traditional data sources, such as cross-sectional census data or aggregate visa statistics, are often unable to link pre-migration educational human capital to post-migration labor market outcomes. To overcome this limitation, we utilize a comprehensive dataset of education and employment histories sourced from Revelio Labs. Revelio Labs aggregates and standardizes public LinkedIn profiles, providing granular, longitudinal information on the educational and professional trajectories of millions of professionals worldwide. The dataset allows us to observe detailed, timestamped histories for each individual. For educational spells, the data include raw and standardized institution names, degree types, fields of study, and precise dates of attendance. For employment spells, we observe job titles, company names, start and end dates, and granular geographic locations.

3.2 Sample Construction

We construct our primary analysis sample by restricting our focus to Indians pursuing master’s studies abroad. This restriction is motivated by two key factors. First, as previously discussed, India represents one of the largest and most strategically responsive source countries for international students globally. Second, from a data reliability perspective, Indian professionals exhibit substantially higher adoption rates of LinkedIn compared to individuals from other major source countries, such as China, thereby ensuring more comprehensive data coverage and representativeness for our analysis.

Because self-reported nationality is not available in LinkedIn data, we proxy for Indian origin using the location of an individual’s undergraduate institution. Specifically, we restrict the sample to users whose most recent bachelor’s degree was granted by an Indian institution and who subsequently report enrollment in a foreign master’s program. We further limit the sample to cohorts commencing their first foreign master’s degree between 2008 and 2013. We additionally exclude individuals pursuing a master’s degree en route to a Ph.D. by dropping observations where the reported start year of a master’s program perfectly coincides with the start year of a doctoral program. These sequential restrictions yield a final primary analysis sample of 94,606 unique individuals. [LinkedIn Coverage Rate](#)

3.3 Variable Construction

Undergraduate Major: A central feature of global high-skilled migration is the disproportionate representation of students in technical fields. We construct a binary indicator equal to one if the student’s undergraduate field of study is classified within a Science, Technology, Engineering, or Mathematics (STEM) discipline. This classification is based on the standardized major text (e.g., “BEng,” “Computer Science,” “Electrical Engineering”) reported for their last Indian bachelor’s field and degree. We exclude students with undergraduate degrees in medicine, law, agriculture, or architecture as career trajectories in these regulated professions differ systematically from those in standard STEM and non-STEM fields.

Undergraduate Institution Quality: To proxy for students’ baseline ability, we categorize individuals based on the quality of their undergraduate institution. Our primary measure utilizes both the overall and discipline-specific rankings from the National Institutional Ranking Framework (NIRF). Introduced in 2015 by the Indian Ministry of Education, the NIRF provides a standardized, annual assessment of higher education institutions across the country (see Appendix B for detailed ranking criteria). Using these data, we assign STEM students, accounting for XX% of our sample, to one of three ability tiers based on the rank of their bachelor’s institution. Specifically, Tier 1 comprises Institutes of National Importance (INIs)¹⁴ and the top-50 colleges in either the NIRF Overall or NIRF Engineering rankings; Tier 2 comprises remaining colleges ranked by either of these systems; and Tier 3 comprises students whose undergraduate institution is unranked.

For non-STEM students, we do not further classify bachelor institutions, for three reasons. First, non-STEM students are fewer than STEM students, so a finer partition would reduce the precision of the estimates. Second, no credible ranking of undergraduate education exists for non-STEM fields. The NIRF does publish a management ranking, but it evaluates postgraduate management programs rather than undergraduate commerce education, and several of the most selective undergraduate business colleges (e.g., Narsee Monjee and Jai Hind College) do not appear in it. Third, some Indian students only report the degree-awarding university, whereas admission and teaching take place at the level of constituent or affiliated colleges whose selectivity varies widely within the same university. For example, a profile listing “University of Delhi” is consistent with St. Stephen’s College or Shri Ram College of Commerce – among the most selective undergraduate institutions in the country – and equally with a lower-tier affiliated college, which awards the same degree title through distance study. The self-reported LinkedIn education profile does not identify which. This ambiguity is more consequential for non-STEM fields. A large share of STEM graduates

¹⁴XX: introduce INI here

hold degrees from standalone technical institutions, like the Indian Institutes of Technology (IIT), which appear on the profile under their own name and are ranked individually in the engineering table of the NIRF. Reporting affiliating universities is therefore more common among Non-STEM individuals, where the profile records the umbrella institution rather than the college.

Undergraduate Attainment: Institutional tier captures selectivity at the point of undergraduate admission, but it does not capture students’ academic performance once enrolled. We therefore construct a separate individual-level indicator of undergraduate academic attainment, equal to one if a student’s LinkedIn bachelor education profile reports distinction on any of three self-reported dimensions. The first is an honours designation on the degree title. The second is an explicit distinction marker, including first class, first division, distinction, a university rank or institute medal, *summa cum laude*, valedictorian, or dean’s list. The third is a numeric grade at or above the conventional distinction threshold on the reported scale.¹⁵

First Job Placement: To capture early-career labor market outcomes, we identify the first employment or placement spell that commences after the completion of the foreign master’s degree. We extract the location, job title, and company name of this initial placement, allowing us to identify the country of first employment and assess whether the student entered the host country’s labor market, returned to India, or migrated to a third country. We further classify each placement along three dimensions.

Job Level. We classify each job into one of three tiers: elite high-skill, regular high-skill, and low-skill occupations. The classification rests primarily on the O*NET Job Zone associated with the occupation code that Revelio Labs imputes from job title and employer. O*NET Job Zones are a five-category classification of occupations by the education, experience, and training they typically require, ranging from Zone 1 (little or no preparation) to Zone 5 (extensive preparation). We classify Zone 5 occupations, most of which typically require graduate school education, as elite high-skilled. The most common job titles in this tier that appear in our data are XX.

We draw the boundary between the remaining two tiers at the skill threshold that the policy itself applies. Post-reform Tier 2 (General) sponsorship required the occupation to sit at UK NQF Level 6 or above, so all Zone 4 occupations, together with those Zone 3 occupations that map to Level 6 or above on the NQF occupation list, are classified as regular high-skill occupations. Some of the most common titles in this tier are XX. We

¹⁵These thresholds are 8.5 on a ten-point scale, 3.7 on a four-point scale, 75 on a percentage or hundred-point scale, and 5.25 on a seven-point scale. Where a grade point average is reported without a stated denominator, we assign the scale by magnitude and leave ambiguous values unassigned; we exclude five-point scales, on which marks in several grading systems run in reverse order.

classify Zone 1 and Zone 2 occupations, together with Zone 3 occupations below NQF Level 6, as low-skill occupations.

Wages. Wages are not directly observed in LinkedIn data; we use wage estimates imputed by Revelio Labs based on job titles, geographic locations, and other employer characteristics.¹⁶

A non-trivial share of master’s graduates immediately enter another educational program rather than the labor market. We separately classify students who transition from the foreign master’s program directly into a doctoral program (“PhD”) or into a second master’s program (“Second Master’s”). These students do not appear as workers in our placement classification but are tracked as separate post-graduation outcomes in the structural model.

3.4 Summary Statistics

Table 1: Summary Statistics (Baseline, Pre-reform)

	All	STEM	Non-STEM
Count	43,825	35,184	8,641
Female	27.82%	24.96%	39.49%
Tier 1 Bachelor	12.67%	13.92%	7.56%
Tier 2 Bachelor	9.54%	10.58%	5.32%
UK Master	25.04%	21.09%	41.08%
US Master	51.54%	57.19%	28.54%
IN Work	17.35%	14.71%	28.12%
Abroad Work	63.86%	66.65%	52.47%
Graduate-Level Work	72.80%	74.93%	64.11%
Non-Graduate-Level Work	8.75%	6.80%	16.70%
PhD	7.38%	7.91%	5.24%

Table 1 reports summary statistics for our sample of 43,825 Indian international master’s students enrolled between 2008 and 2010 before the reform, separately for STEM and non-STEM students. The sample is heavily skewed toward STEM fields, which account for 81% of students. Women are a minority of the sample (28.9%) but represent a substantially larger share of non-STEM students (41.6%) than of STEM students (25.9%). Approximately 25% of students completed their bachelor’s degree at a Tier 1 or Tier 2 institution under our NIRF-based ability classification, with the remainder (75%) drawn from unranked institutions.

¹⁶Some justification or detailed document of Revelio Labs here...

Master's destination patterns differ markedly by field. STEM students predominantly chose the United States (59.2%) over the United Kingdom (15.6%), while non-STEM students were nearly evenly split between the two destinations (UK: 36.6%; US: 30.2%). As for post-graduation, 65.3% of students secured employment outside India, with 69.2% of all employed students entering graduate-level (RQF-6) occupations. STEM students were both more likely to work abroad (68.5% versus 51.3%) and more likely to enter graduate-level positions (70.8% versus 62.6%). A small but non-negligible share of students continued to a doctoral program (6.8% overall), with PhD continuation more common among STEM students (7.3%) than non-STEM students (4.9%).

3.4.1 Descriptive Evidence about the Policy Shock

Before proceeding to the formal econometric and structural analysis, we present descriptive evidence highlighting the stark aggregate and compositional shifts triggered by the 2012 U.K. visa reform. The raw data reveal three stylized facts regarding the policy's impact on global spatial sorting, search frictions, and the composition of retained talent.

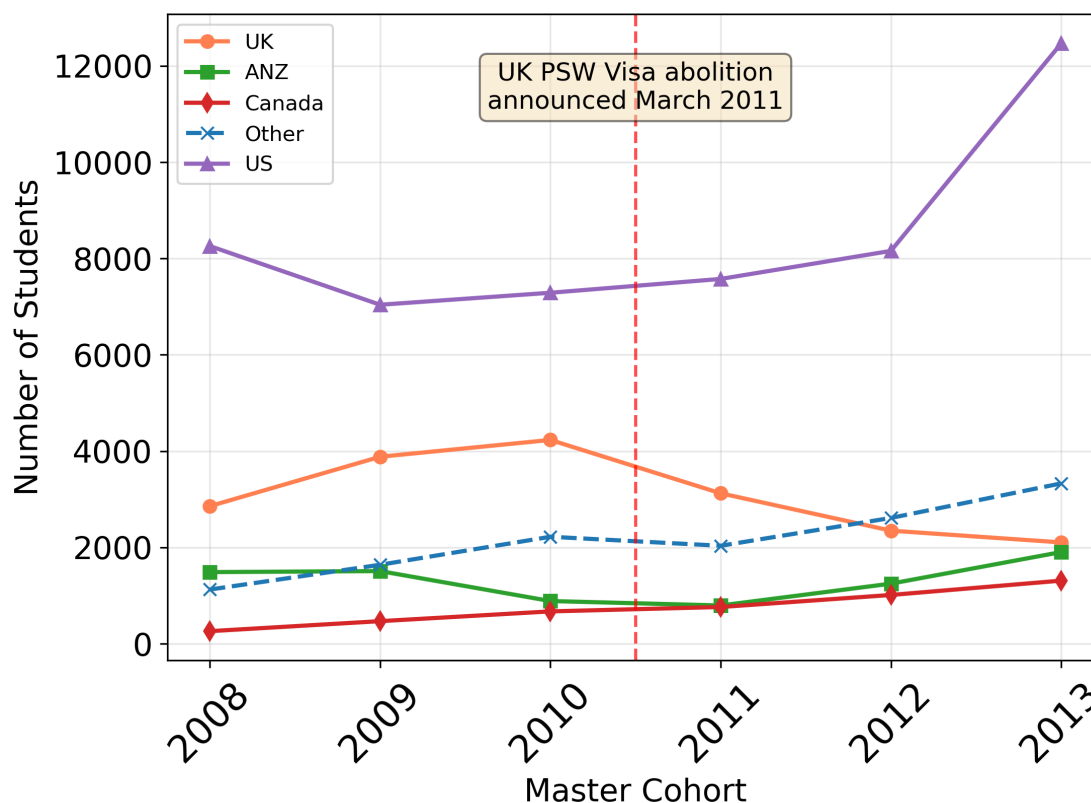


Figure 2: Indian Foreign Master's Students by Year

Fact 1: A Severe Deterrence Effect on Enrollment and Retention.

Following the announcement and implementation of the reform, the U.K. experienced a sharp, absolute decline in its ability to attract and retain global talent. As illustrated in Figure 2, master’s enrollment of Indian students in the U.K. plummeted starting with the treated 2011 and 2012 cohorts. In contrast, enrollment trends in competing destinations—such as the United States and the Rest of the World (ROW)—remained relatively flat or exhibited slight growth during the same period, suggesting that the U.K. decline was driven by destination-specific policy rather than origin-country macroeconomic shocks.

Fact 2: Compositional Change

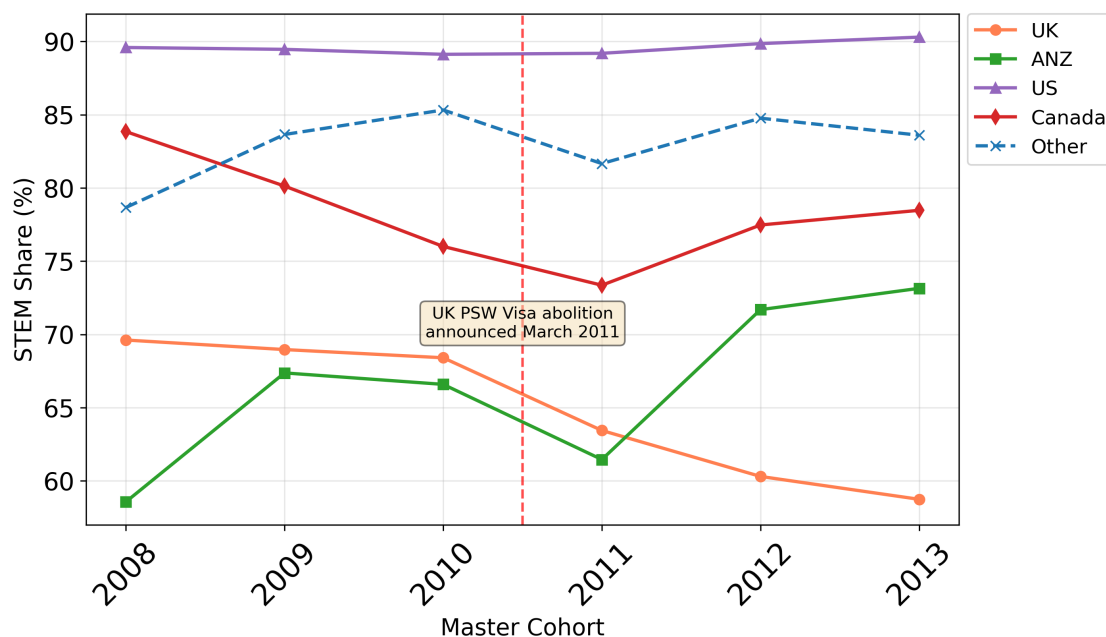


Figure 3: STEM Share By Destination Country

Two compositional shifts accompanied the enrollment decline. First, the field mix of UK-bound Indian students shifted away from STEM. Figure 3 plots the STEM share of Indian master’s students at each destination over the sample period. Among UK-bound students, the STEM share fell from approximately 70% in the 2010 cohort to 60% in the 2013 cohort — a sharper decline than at any other destination — indicating that STEM students were disproportionately deterred from the UK after the reform. Second, the ability composition of UK-bound students shifted upward. Figures 4 and 5 examine the bachelor-tier composition of Indian students at each destination. Among UK master’s students, the share enrolled from Tier 1 institutions rose from approximately 8% in the 2008 cohort to 16% in the 2013 cohort for STEM students, and from 11% to 17% for non-STEM students; the combined Tier 1 and Tier 2 share rose correspondingly. Figure 5 shows that this shift toward elite-institution students occurred at the UK more sharply than at other destinations

during the same period, although the US and "Other" destinations also experienced gradual upward trends in elite-institution share. Both patterns are consistent with a policy that disproportionately deterred lower-tier and non-STEM students, leaving the surviving UK student body more selectively drawn from elite STEM-track candidates. We caution that these descriptive shifts cannot, on their own, distinguish between selection on the demand side (students with stronger outside options choose the UK less) and selection on the supply side (UK universities admit better students as program demand falls).

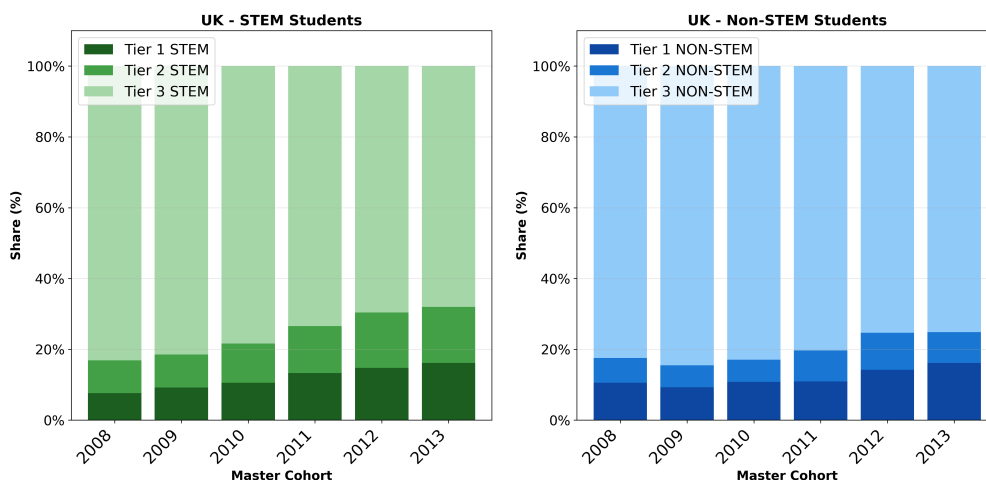


Figure 4: Indian Bachelor Tier Distribution for UK Master

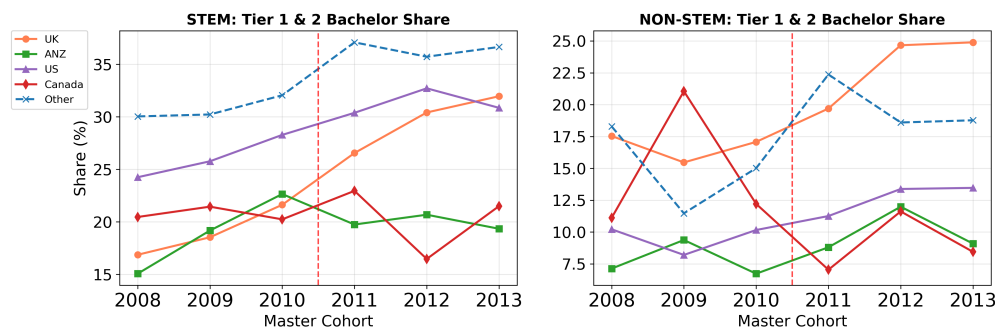


Figure 5: Elite & Good Bachelor Institution Share By Destination Country

Fact 3: Employment.

Figure 6 shows the distribution of post-graduation outcomes by cohort. The share of UK master's graduates whose first job was in the UK (the "Work in Host" category) fell from approximately 32% in the 2008 cohort to 16% in the 2011 cohort and remained at that level through the 2013 cohort. The share returning to India for their first job rose correspondingly, from approximately 38% to 45%. The share migrating to a third country also rose modestly. Figure 7 conditions on students with any observed work outcome and

decomposes the host-country employment by job skill level. Both graduate-level and non-graduate-level employment in the UK fell as a share of total work outcomes; the graduate-level share fell from approximately 27% in 2008 to 14% in 2011, while the non-graduate-level share fell from 11% to 5%. The reform reduced UK retention across both skill categories, not only the low-skilled category that the policy targeted.

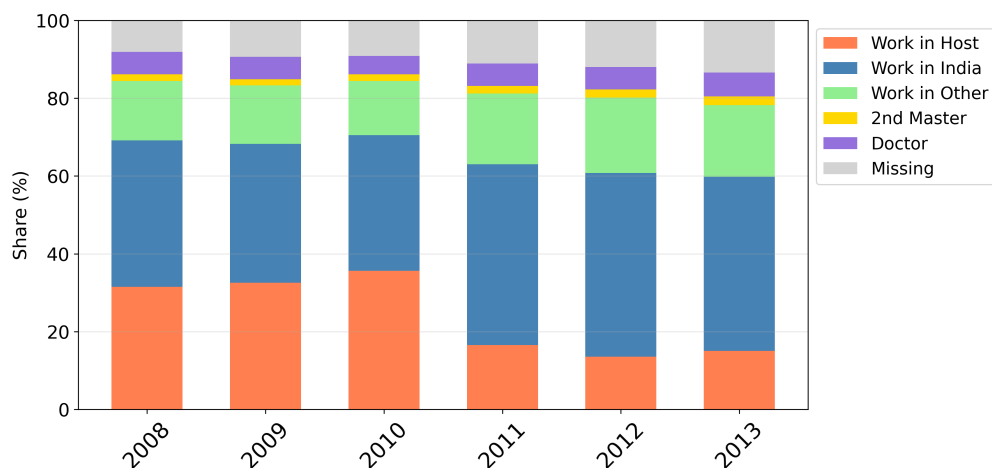


Figure 6: Post-Graduation Placement for UK Master

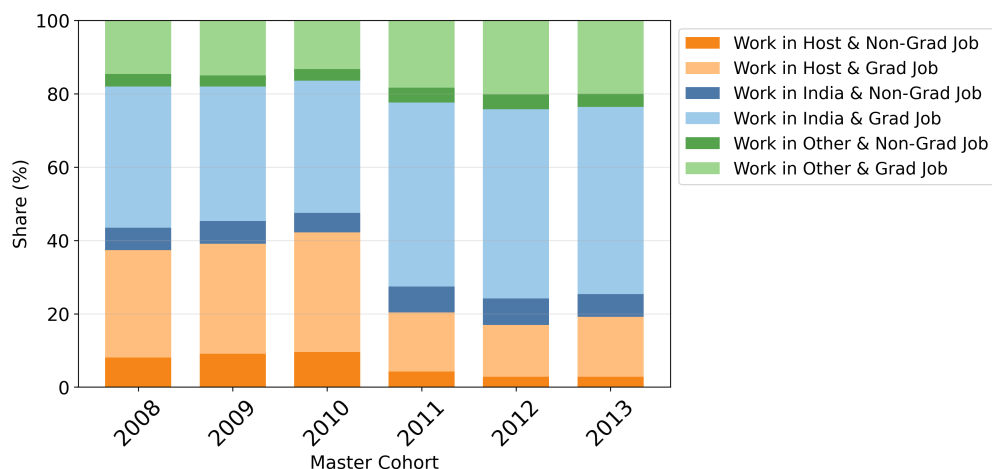


Figure 7: Post-Graduation Placement for UK Master, Cond' on Work

Figure 8 summarizes the levels of UK first-job employment among Indian master's graduates by cohort. Graduate-level UK employment peaked at 2,756 in the 2010 cohort and fell to 1,386 by the 2013 cohort, a 50% decline. Non-graduate-level employment fell from 785 to 248 over the same period, a 68% decline. While the proportional decline in non-graduate-level employment was larger, consistent with the policy's stated objective of restricting low-skilled

stay, the absolute decline in graduate-level employment was substantially larger in magnitude (a loss of 1,370 graduate-level jobs versus 537 non-graduate-level jobs). This pattern is the central descriptive fact that motivates our structural analysis: the policy reduced low-skilled stay, but it also reduced graduate-level retention by a comparable absolute amount. Whether this graduate-level decline reflects deterrence at the education margin (fewer UK enrollees) or increased frictions at the labor-market margin (UK enrollees less able to secure graduate-level employment) is the question we take up formally in Sections [X] and [X].

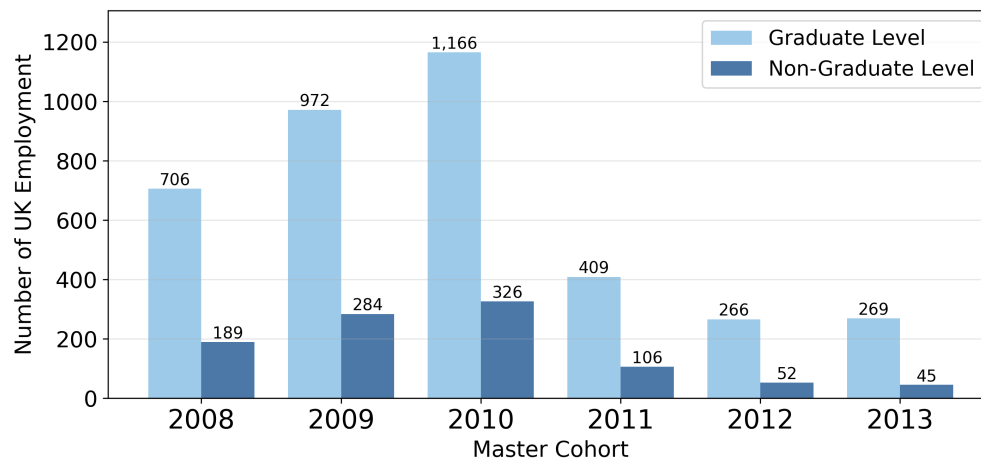


Figure 8: UK first-job employment by master cohort (2008-2013)

4 Quasi-Experimental Evidence

To identify the impact of the 2011 UK immigration policy change on international student flows, we employ an event-study framework that leverages cross-sectional variation in Indian states' historical exposure to the UK labor market. Our identification strategy rests on the intuition that students from Indian states with a higher historical propensity to remain in the UK for work are more sensitive to the restriction of this pathway. Consequently, we expect these highly exposed states to experience a disproportionately larger decline in enrollment following the policy implementation. As is common in empirical designs lacking a pure, unexposed control group (e.g., Dustmann et al., 2022), our identification strategy requires a linear dose-response assumption in addition to the standard parallel trends assumption. Specifically, we assume that the treatment effect on the outcome variable—the log of enrollment—is strictly linear with respect to the continuous measure of exposure, which in our context is the historical stay rate. The detailed econometric assumptions is written in Appendix D.

Specifically, we specify our primary event-study model as follows :

$$\ln(Y_{st}) = \sum_{\tau=2006}^{2014} \beta_{\tau}(\text{StayRate}_s \times \mathbf{1}\{t = \tau\}) + X'_{st}\gamma + \alpha_s + \delta_t + \epsilon_{st} \quad (1)$$

where:

- Y_{st} is the number of students from state s enrolling in a Master's program in the United Kingdom in year t .
- StayRate_s represents the treatment intensity, defined as the mean probability that a student from state s remained in the UK for work following their studies during the pre-reform period (prior to 2010).¹⁷
- α_s and δ_t are state and year fixed effects, respectively. These account for time-invariant state characteristics and aggregate shocks to the demand for international education.
- X_{st} is a vector of time-varying state-level controls, specifically the natural logarithm of Gross State Domestic Product (GSDP) to control for potentially non-linear local economic drivers of migration.
- ϵ_{st} is the idiosyncratic error term.

¹⁷To mitigate small-sample bias, we exclude states with fewer than 50 U.K. master's degree observations prior to 2010. This restriction yields a final analytical sample of 21 states.

The coefficients of interest, β_τ , measure the differential change in enrollment for states with varying levels of historical reliance on the UK post-study work pathway. To satisfy the parallel trends assumption, we expect β_τ to be statistically indistinguishable from zero for $\tau < 2011$. For $\tau \geq 2011$, a negative and significant β_τ would indicate that students from states more reliant on the PSW pathway were more responsive to the restriction.

4.1 Variation in Historical Retention Rate

As shown in Figure 9, Pre-2010 stay rates vary substantially across Indian states, ranging from approximately 20% in low-stay states to approximately 45% in high-stay states. This variation reflects systematic institutional differences across Indian states in the role of foreign work as a post-graduation outcome. Two regions display particularly high pre-policy UK stay rates. In Northwest India — Punjab and Gujarat in particular — long-standing UK diaspora networks lower the search and assimilation costs of remaining in the UK for work, sustaining a high propensity to stay among students from these states. In South India — particularly Karnataka and Kerala — high stay rates reflect labor market complementarity rather than diaspora ties: these states are the source of a large share of India’s outsourcing and IT services workforce, and skilled migration to the UK technology sector is a salient post-graduation pathway. By contrast, students from the central and northern “Hindi Belt” states display markedly lower pre-policy UK stay rates. In these states, family businesses, civil-service careers, and marriage-market considerations make returning to India the more salient post-graduation option, independent of the labor market access offered by host countries. Crucially, these regional patterns reflect characteristics of the Indian states themselves, less related to the features of the 2011 UK reform.

4.2 Main Result

Figure 10 plots the event-study coefficients (β_τ), which empirically validate our identification strategy: prior to the 2011 policy shock, the estimated coefficients are statistically indistinguishable from zero, supporting the parallel trends assumption. Following the March 2011 abolition of the Tier 1 Post-Study Work (PSW) visa, we observe a precipitous and statistically significant contraction in UK enrollment among highly exposed states. The magnitude of the crowding-out effect intensifies over time, with the coefficient dropping to approximately -5 log points by 2012, underscoring the high elasticity of international student flows to host-country work authorization policies.

We further condense this variation into a generalized difference-in-differences (DiD) specification:

Pre-2010 UK Stay Rate by Indian State

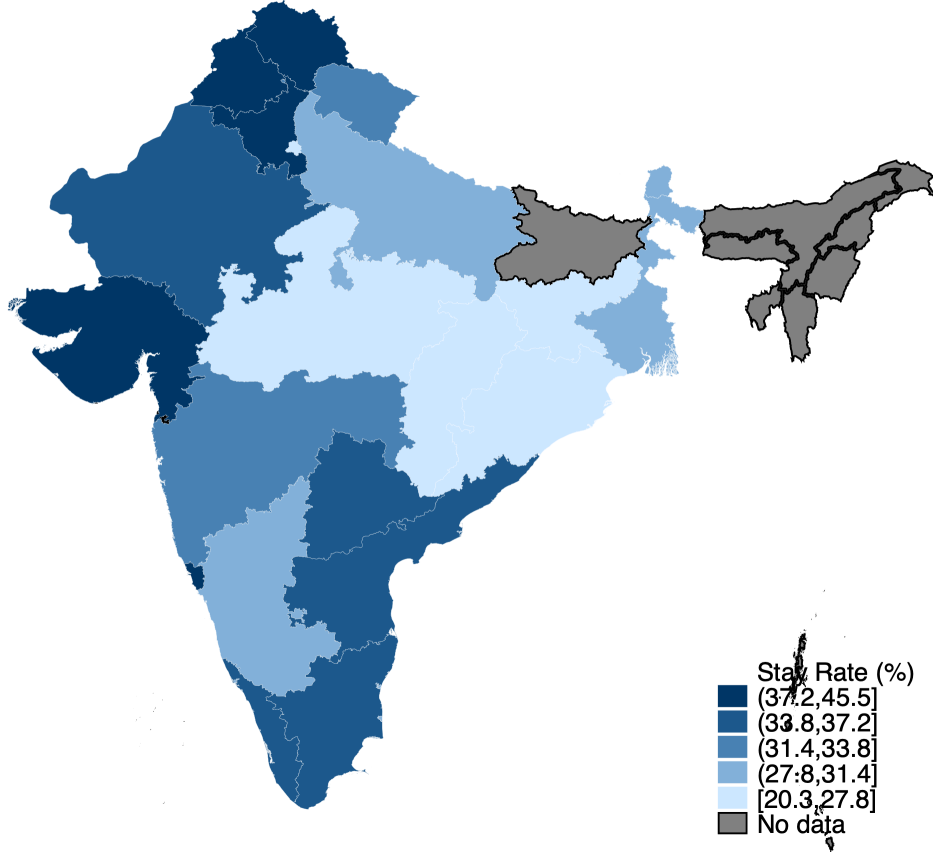


Figure 9: Pre-Reform Stay Rate Across States

$$\ln(Y_{st}) = \alpha_s + \delta_t + \beta(\text{StayRate}_s \times \text{Post}_t) + \Gamma X_{st} + \epsilon_{st} \quad (2)$$

where Post_t is an indicator variable equal to one for the years 2011–2014. In all specifications, we cluster standard errors at the state level to account for arbitrary serial correlation within states.

The estimated coefficient on $\text{StayRate}_s \times \text{Post}_t$ in the pooled specification is $\hat{\beta} = -2.29$, implying a substantial reduction in UK master's enrollments. To translate this estimate into an aggregate magnitude, we apply the state-specific treatment intensity to pre-policy enrollment counts:

$$\Delta Y = \frac{\sum_s N_s^{UK} \exp(\hat{\beta} \cdot \text{StayRate}_s)}{\sum_s N_s^{UK}} - 1 = -0.53,$$

indicating an approximate 53% decline in UK master's enrollment of Indian students attributable to the 2011 reform.

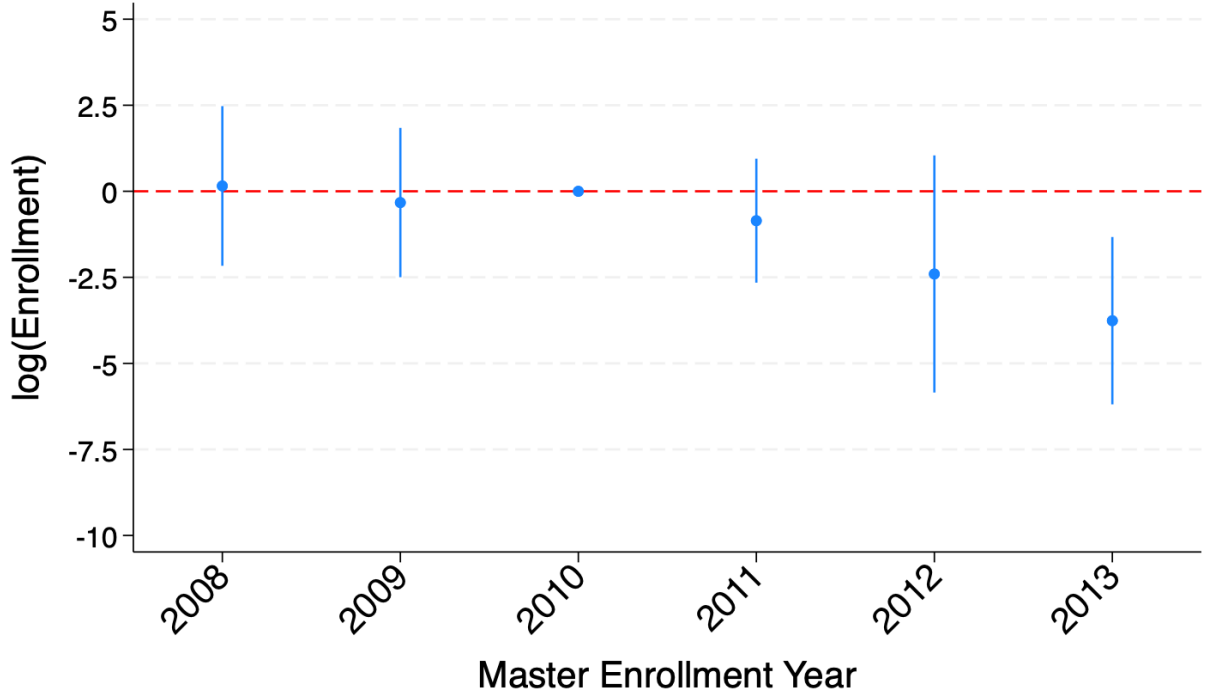


Figure 10: Event-Study UK Enrollment

The corresponding fiscal magnitude is large. According to HESA aggregate data, approximately 20,000 Indian students enrolled in UK taught master’s programs in the 2010 cohort, paying average tuition of approximately £10,431 per year. Applying the estimated 53% reduction to the pre-policy enrollment level — and noting that UK taught master’s programs typically last one year, so each cohort’s tuition revenue is realized within a single year — we estimate that the reform reduced UK universities’ annual revenue from Indian master’s students by approximately £110 million. We emphasize that this figure represents only the direct revenue effect from Indian students, who are one source population among many; the total revenue effect across all source countries would be substantially larger if the reform’s effects on non-Indian students are of similar magnitudes.

4.3 Heterogeneity

Table 2 presents the Diff-in-Diff estimates for different subsamples. The estimated effect is concentrated among STEM students: $\hat{\beta}^{STEM} = -3.231$), statistically significant at the 1% level. The corresponding non-STEM estimate is $\hat{\beta}^{Non-STEM} = -1.822$, negative in sign but not statistically distinguishable from zero at conventional levels. Restricting attention to Tier 3 students, those whose bachelor’s degrees were earned at NIRF-unranked institutions,

yields coefficients of -2.634 for STEM and -1.884 for non-STEM. The Tier 3 STEM estimate is smaller in absolute value than the pooled STEM estimate, indicating that the response is larger for high ability STEM students.

The heterogeneous response is consistent with pre-policy differences in the UK's role as a post-graduation destination across fields. In the 2008–2010 cohorts, 35.7% of UK-educated Indian STEM students entered the UK labor market after graduation, compared with 28.6% of non-STEM students. STEM Students with higher baseline reliance on the UK labor-market route should respond more strongly to a policy that removes that route in the education destination-choice margin. The descriptive gap in pre-policy stay rates helps explain the differential reduced-form response we estimate.

Table 2: Effect of Treatment on Subgroups

	STEM	Non-STEM	STEM Tier 3	Non-STEM Tier 3
Stay Rate $\times$ Post	-3.231*** (0.809)	-1.822 (1.072)	-2.634* (1.364)	-1.884 (1.360)
State FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Cluster by State	Yes	Yes	Yes	Yes
Observations	132	96	108	90

Standard errors in parentheses, clustered at state level.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

5 Model

Our model features in the joint dynamic decision of global master country choice and the immigration choice post-graduation, acknowledging staying in the master host country, earning the wage premium in the developed country is an important incentive in the international master education.

5.1 Setup

We model each individual i 's decisions as a two-stage sequential discrete choice problem. Individual i is characterized by a vector of pre-master's state variables $\Omega_i = (A_i, O_i, t_i)$, defined as follows. A_i indexes the individual's bachelor's-degree characteristics, given by the Cartesian product of undergraduate field (STEM or non-STEM) and three institutional ability tiers defined in Section 3, yielding $|A_i| = 6$ types in total. A_i governs both the individual's preferences over master's destinations and post-graduation alternatives, and the wage returns associated with each labor-market outcome. O_i denotes the Indian state in which the individual completed their bachelor's degree. O_i enters the model through state-level GDP, which serves as a proxy for the financial resources available to the individual for financing master's education. In addition, students from different states might have heterogeneous tastes for education and immigration, as is shown in the quasi-experimental evidence. Finally, t_i indexes the cohort year of master's enrollment, taking values in $\{2008, 2009, \dots, 2013\}$. The cohort indicator determines the policy environment the individual faces (pre-reform for $t_i \leq 2010$ and post-reform for $t_i \geq 2011$) and also pins down the macroeconomic conditions that shape the individual's wage expectations at the time of decision-making.

In the first stage, the individual chooses a master's destination j^{MA} from the set ¹⁸

$$\mathcal{J}_1 = \{\text{UK, US, Canada, ANZ (Australia and New Zealand), Europe (Low), Europe (High)}\}$$

The master destination "Europe (Low)" comprises countries charging very low tuition during the period, such as France and Germany; "Europe (High)" comprises countries charging a non-negligible amount of tuition for master's programs, such as Ireland and Sweden.¹⁹

In the second stage, the individual selects a post-master's outcome j^W from a choice set $\mathcal{J}_2(j^{MA})$ that depends on the first-stage master's destination.

A student who chooses the UK at Stage 1, $j^{MA} = \text{UK}$, faces sixteen Stage-2 alternatives in total. Out of the sixteen choices, fourteen working choices include: three in the UK, one

¹⁸These destinations account for XXX share of master's education.

¹⁹Swedish master's programs were free before 2010 but charged substantial fees after 2011.

in India, and ten in a third country. Within the UK, she can take an elite high-skill job, a regular high-skill job, or a low-skill job. The two high-skill alternatives are realized stochastically, and she can take one only if it is realized. Realization is alternative-specific, so availability of one does not imply availability of the other. We will elaborate this more in the next section. Outside the UK, she can return to India to work or work in the third country (five groups), including the United States, Canada, Australia and New Zealand (ANZ), Europe, or the rest of the world. For each third country group, she can choose a high or low-skill job. The individual also faces two continued-education alternatives: enrolling in a doctoral program or in a second master's program. Together with the fourteen employment alternatives, these complete the Stage-2 choice set J_2 .²⁰

A student who chooses a master's degree outside the UK, $j^{MA} \neq \text{UK}$, faces a choice set similar to that of a UK master's student except that we do not distinguish the two types of high-skill employment as we do for UK master's students. Upon graduation the individual faces fifteen alternatives: high- and low-skill employment in the host country (2), high- and low-skill employment in each of five third countries (2×5), return to India (1), and the two continued-education alternatives described above (2).²¹

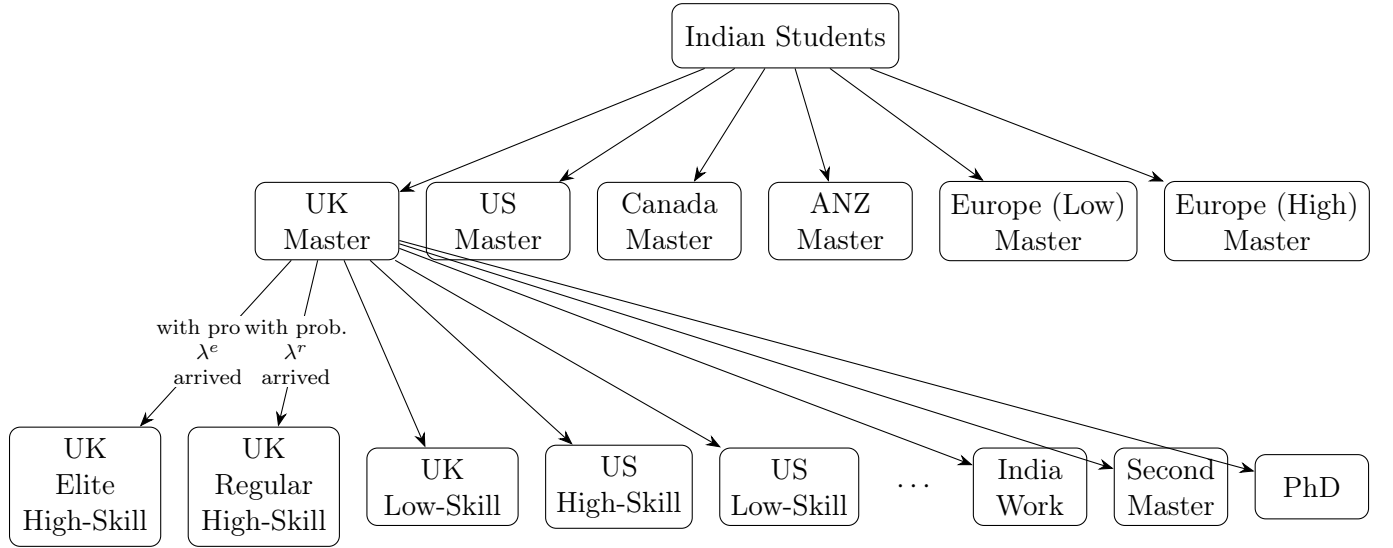


Figure 11: Two-stage sequential choice structure.

²⁰We treat each continued-education alternative as a single option and do not index it by the country in which the program is undertaken. These alternatives account for approximately XX% of observed post-master's outcomes; disaggregating them by location would sacrifice precision in the corresponding preference parameters.

²¹For a student who enrolls in a US master's program, the five third countries are Canada, ANZ, the UK, Europe, and the rest of the world. For UK high-skill employment, we group the elite and regular high-skill alternatives together.

5.2 Stochastic Choice Set and Utility

5.2.1 Stage 2: Work Location Choice

Since our policy variation comes from the post-study work environment faced by UK master's graduates, we treat the work utility of a UK master's graduate seeking UK employment differently from the others. Specifically, UK low-skilled jobs appear in the choice set before the policy but are removed entirely thereafter.²² UK high-skilled jobs, by contrast, appear stochastically in the choice set of students completing a UK master's degree. We denote this probability by λ^{pre} for cohorts $t \leq 2010$ and λ^{post} for cohorts $t \geq 2011$. We expect $\lambda^{post} < \lambda^{pre}$, capturing the policy's effect on hiring frictions (a shortened job-search window, employers' reduced willingness or increased difficulty in sponsoring visas, and so on). For UK master's students with a UK high-skilled or low-skilled job in their choice set, the utility of UK work alternatives is

$$U_{UK,q,t}^W(j^{MA} = UK, A) = \underbrace{\beta^W \mathbb{E}[\ln w_{UK,q,t}(UK, A)]}_{\text{Pecuniary Utility}} + \underbrace{\eta(UK, q; UK, A, O)}_{\text{Amenity}} + \epsilon_{j^w_{qt}}^W$$

where $\mathbb{E}[\ln w_{UK,q,t}(UK, A)]$ is the individual's expected log wage in the UK for a first job at skill level q . The belief is constructed from observed cell-level fixed effects, ability shifters, master's-match shifters, and the persistent component of macroeconomic wage shocks; we describe wage-belief formation in Section [X]. The amenity term $\eta(UK, q; UK, A, O)$ represents the non-pecuniary factors of UK employment for a UK master of type A from origin O entering a job of type q , such as career progression, environment, and home preference relative to the origin (i.e., India). The ϵ shocks are idiosyncratic random shocks.

For the remaining post-graduation employment alternatives $(j^W) \in \mathcal{J}_2$, and for master's destinations other than the UK, we assume that the indirect utility of individual i choosing work alternative j^W , conditional on the Stage-1 master's destination j^{MA} and individual type A , is

$$U_{j^w_t}^W(j^{MA}, A) = \underbrace{\beta^W \mathbb{E}[\ln w_{j^w_t}(j^{MA}, A)]}_{\text{Pecuniary Utility}} + \underbrace{\xi(j^W; j^{MA}, A)}_{\text{Non-wage Utility}} + \epsilon_{j^w_t}^W$$

where $\mathbb{E}[\ln w_{j^w_t}(j^{MA}, A)]$ is the expected log wage in work choice j^W for an individual in cohort t of type A who obtained a master's in j^{MA} . The non-wage utility ξ differs in meaning from the amenity term η : whereas η captures amenities alone, ξ is a composite of both ameni-

²²The small sample of UK low-skilled employment spells shown in Figure 8 are excluded from the estimation.

ties and job-search frictions. The specification of ξ takes a piecewise form that distinguishes three regimes, with the $\{\text{India}\}$ alternative serving as the reference category ($\xi^{\text{India}} \equiv 0$). Every utility parameter is therefore interpreted relative to this reference category. The detailed specification of ξ is illustrated below.

First, when the individual works in the same country as their master's degree, the non-pecuniary utility is given by $\xi^{\text{Stay}}[q, j^{MA}, A]$, which is allowed to vary fully by job skill, host country and individual types. Second, when the individual works in a third country different from both their master's education country and home country, the non-pecuniary utility depends on the final work country $\xi^{\text{Third}}[j^W]$, which depends largely on immigration or work right visa policy for global talents. Combining the two cases together:

$$\xi(j^W, q; j^{MA}, A, t) = \begin{cases} \xi^{\text{Stay}}[q, j^{MA}, A] & \text{if } j^W = j^{MA} \in \{\text{US}, \text{Canada}, \text{ANZ}, \text{Europe}\} \\ \xi^{\text{Third}}[j^W] & \text{if } j^W \neq j^{MA}, j^W \in \mathcal{J}^{MA} \cup \{\text{Other}\} \end{cases}$$

$\epsilon_{j^W t}^W$ is an i.i.d. Type-I extreme-value shock, yielding a conditional logit choice probability over $\mathcal{J}_2^{\text{Work}}$.

5.2.2 Stage 2: PhD and Second Master's Choice

For the two continued-education alternatives in $\mathcal{J}_2$, we model both alternatives through preference parameters $\xi^{\text{PhD}}[j^{MA}, A]$ and $\xi^{\text{2ndMA}}[j^{MA}, A]$, which absorb all systematic determinants of the continued-education choice conditional on the master's destination and individual type. Thus, the indirect utility of individual i is given by

$$\begin{aligned} U^{\text{PhD}}(j^{MA}, A) &= \xi^{\text{PhD}}[j^{MA}, A] + \epsilon^{\text{PhD}}, \\ U^{\text{2ndMA}}(j^{MA}, A) &= \xi^{\text{2ndMA}}[j^{MA}, A] + \epsilon^{\text{2ndMA}}, \end{aligned}$$

where the shocks ϵ^{PhD} and ϵ^{2ndMA} are i.i.d. Type-I extreme-value, distributed independently of the work-alternative shocks $\epsilon_{ij^W t}^W$ defined above.

5.2.3 Stage 1: Master's Destination Choice

In Stage 1, the individual chooses a master's destination $j^{MA} \in \mathcal{J}_1$. The indirect utility of choosing destination j^{MA} at cohort t , conditional on ability type A_i and origin state O_i , is

given by

$$\begin{aligned}
U_{j^{MA}t}^{MA}(A, O_i) = & \underbrace{(\beta_1 + \beta_2 W_t(O_i)) P_{j^{MA},t} \chi_{j^{MA},t}}_{\text{Tuition cost}} \\
& + \underbrace{\varphi_0[j^{MA}, A_i]}_{\text{Education Consumption Value}} \\
& + \underbrace{\delta(j^{MA}) \cdot IV[j^{MA}; A_i, O_i, t]}_{\text{Continuation Value}} + \epsilon_{ij^{MA}t}^{MA},
\end{aligned}$$

where $P_{j^{MA},t}$ is the average tuition charged to international master's students at destination j^{MA} in cohort year t , expressed in destination-country currency. For each destination, we use the cohort-specific mean tuition across universities / countries within the composite destinations. Because master's program length differs across destinations, $P_{j^{MA},t}$ measures total program tuition rather than an annual rate: we multiply annual tuition by the destination's average program length.²³ $\chi_{j^{MA},t}$ is the annual-average nominal exchange rate between the Indian rupee and the currency of destination j^{MA} in year t , converting destination-country tuition into Indian rupees. $W_t(O_i)$ is the per-capita GDP of the individual's origin state O_i at cohort year t , proxying for the financial resources available to the individual's household. β_1 is the baseline tuition sensitivity for students from extremely poor regions, and β_2 is the wealth-mediated attenuation: $\beta_2 > 0$ implies that students from wealthier origin states are less deterred by tuition.

$\varphi_0[j^{MA}, A_i]$ is a destination-and-type-specific intercept capturing the non-pecuniary consumption value of obtaining a master's degree at destination j^{MA} for a student of type A_i . This includes non-wage amenities of the destination (e.g. cultural fit) and the returns not related to the labor market. We treat φ_0 as time-invariant during our sample period and normalize $\varphi_0[UK, A] \equiv 0$ for each A as the reference category. Therefore, $\varphi_0[j^{MA}, A_i]$ should be interpreted as for a student with type A_i , how much she places on education consumption value relative to the corresponding UK education.

$IV[j^{MA}; A_i, O_i, t]$ is the inclusive value of the Stage-2 choice problem conditional on having chosen destination j^{MA} . It differs between the UK master's option and other countries' master's options because UK graduates face a stochastic choice set. Specifically, the inclusive

²³Because tuition is observed at the cohort-destination level rather than at the individual-program level, $P_{j^{MA},t}$ implicitly reflects an expectation taken over the set of programs to which an individual might apply, consistent with a sequential decision structure in which the destination is chosen before specific programs are applied to. Appendix E illustrates the details on how to construct tuition variables.

value of a UK master's is

$$\begin{aligned}
IV[j^{MA} = \text{UK}, t \leq 2010] &= \lambda_{\text{pre}} \log \left(\sum_{j^W \in \mathcal{J}_2(\text{UK})} \exp V_2(j^W; \text{UK}, A_i, t_i; \theta) \right) \\
&\quad + (1 - \lambda_{\text{pre}}) \log \left(\sum_{j^W \in \mathcal{J}_2(\text{UK}), j^W \neq (\text{UK}, H)} \exp V_2(j^W; \text{UK}, A_i, t_i; \theta) \right) \\
IV[j^{MA} = \text{UK}, A, t \geq 2011] &= \lambda_{\text{post}} \log \left(\sum_{j^W \in \mathcal{J}_2(\text{UK}), j^W \neq (\text{UK}, L)} \exp V_2(j^W; \text{UK}, A_i, t_i; \theta) \right) \\
&\quad + (1 - \lambda_{\text{post}}) \log \left(\sum_{j^W \in \mathcal{J}_2(\text{UK}), j^W \neq (\text{UK}, H), (\text{UK}, L)} \exp V_2(j^W; \text{UK}, A_i, t_i; \theta) \right)
\end{aligned}$$

Here $V_2(j^W; j^{MA}, A_i, O_i, t_i)$ is the deterministic component of the utility of Stage-2 alternative j^W , realized one period after the master's destination j^{MA} is chosen. The inclusive value summarizes the option value of the post-graduation choice set, spanning expected work, PhD, and second-master's outcomes. For the pre-reform cohort, the UK high-skilled job enters the choice set with probability λ_{pre} , in which case the option value is the log-sum over all alternatives; with probability $1 - \lambda_{\text{pre}}$, the option value excludes the UK high-skilled job. For the post-reform cohort, two changes occur: the UK low-skilled job is removed from the Stage-2 choice set entirely, and the UK high-skilled job's arrival probability changes from λ_{pre} to λ_{post} .

The other master's options do not face a stochastic choice set in the second stage. Their inclusive value is therefore

$$IV[j^{MA}; A_i, O_i, t] = \log \sum_{j^W \in \mathcal{J}_2(j^{MA})} \exp(V_2(j^W; j^{MA}, A_i, O_i, t_i)),$$

under the logit structure.

The discount factor is destination-specific, $\delta(j^{MA}) = 0.95^{T(j^{MA})}$, where $T(j^{MA})$ is the average program length at destination j^{MA} and the annual discount factor is fixed at 0.95. This adjustment matters because programs differ systematically in length, typically one year in some destinations (e.g., the UK) and two in others (e.g., the US). Thus, a student who defers labor-market entry by an extra year discounts the Stage-2 continuation value accordingly. The Stage-1 preference shock $\epsilon_{ij^{MA}t}^{MA}$ is i.i.d. Type-I extreme value, so Stage-1 choice probabilities take the conditional logit form over $\mathcal{J}_1$.

The master's destination choice is driven by three factors: the upfront pecuniary cost of tuition (modulated by household wealth via the origin state's GDP), the destination-specific consumption value of the degree, and the forward-looking option value of the post-graduation choice set. The inclusive-value term is the channel through which the 2011 reform

affects Stage-1 decisions: after the the policy, UK master's students lose the UK low-skilled alternative and face a lower probability of the UK high-skilled alternative in the Stage-2 work set, which lowers the option value and thereby reduces the relative attractiveness of a UK master's destination. Because the reform was announced in March 2011, the 2011 cohort, who enter that September and choose their master's destination shortly beforehand, may not have reached the new steady state for two reasons. First, reapplying elsewhere on short notice is difficult, so this cohort is largely locked into the UK even when it is no longer their first choice. Second, they may not have fully internalized the policy's implications. We therefore model the 2011 cohort as a mixture: with probability τ , they anticipate the UK regime change and adapt their education choice accordingly; with probability $1 - \tau$, they choose as if the pre-reform regime still applied.

5.3 Wage Beliefs and the Stochastic Wage Process

An important component of the model is the specification of students' beliefs over post-graduation wages. Because the master's destination is chosen one or two years before post-graduation wages are realized, students must form expectations over future wage outcomes at the time of the Stage 1 decision. We make these expectations operational by assuming a parsimonious stochastic structure for wages, in which the fixed component is determined by occupational, ability, and master's-match fixed effects, and the stochastic component follows an AR(1) process around these means.

5.3.1 Wage Equation

The realized log wage of individual i in country j^W at occupational skill level q in year t , conditional on the master's destination j_i^{MA} and ability type A_i , is given by

$$\begin{aligned} \ln w_{ij^W qt}(j_i^{MA}, A_i) = & \underbrace{\alpha_{j^W, q}}_{\text{Location-skill FE}} + \underbrace{\sum_{a=2}^{N_A} \delta_A^w(a) \cdot \mathbf{1}\{A_i = a\}}_{\text{Ability shifter}} \\ & + \underbrace{\delta^{MA} \cdot \mathbf{1}\{j^W = j_i^{MA}\}}_{\text{Master's-match premium}} + \underbrace{u_{j^W qt}}_{\text{Macro shock}} + \underbrace{\nu_{ij^W qt}}_{\text{Individual shock}}, \end{aligned}$$

where $\alpha_{j^W, q}$ is a location-by-skill fixed effect, capturing average log wages for occupational skill level q in country j^W . $\delta_A^w(a)$ is an ability-type shifter, with $A_i = 1$ (Tier 3 STEM, the lowest tier) normalized to zero. The full set $\{\delta_A^w(a)\}_{a=2}^6$ captures wage premia for higher ability types relative to the reference. δ^{MA} is a master's-match premium: graduates working in the same country as their master's degree receive an additional log wage shifter, capturing

both network effects and host-country-specific human capital returns. $u_{jw_{qt}}$ is a country-by-skill-by-time macro shock that aggregates business-cycle conditions and other macroeconomic shifters unobserved to the econometrician. We assume $u_{jw_{qt}}$ is observed by students at the time of decision-making but stochastic from the perspective of the wage realization process. $\nu_{ijw_{qt}}$ is an individual-specific wage shock, assumed i.i.d. with mean zero and constant variance σ_ν^2 .

5.3.2 AR(1) Belief Formation

We assume that the “country-by-job-type” macro shock $u_{jw_{qt}}$ evolves according to a first-order autoregressive process:

$$u_{jw_{q,t+1}} = \rho u_{jw_{qt}} + \zeta_{jw_{q,t+1}}, \quad \zeta_{jw_{q,t+1}} \stackrel{\text{iid}}{\sim} \mathcal{N}(0, \sigma_\epsilon^2),$$

where $\rho \in (-1, 1)$ governs the persistence of macroeconomic shocks and σ_ϵ^2 is the innovation variance. The fixed effects $(\alpha_{jw,q}, \delta_A^w, \delta^{MA})$ are time-invariant and assumed known to students.

Students form rational expectations of the wages they will realize upon graduation using the most recent macroeconomic shock they observe at the time of their educational decision. The forecast horizon is set by program length: a student in cohort t graduates at $t+1$ or $t+2$, and we assign the horizon by rounding average program length in the data. UK master’s programs average 1.3 years and are rounded to a one-period horizon; all other destinations are rounded to two periods.

Because the destination-specific shock $u_{jw_{q,t}}$ follows an AR(1) process with persistence ρ , a student forecasting h periods ahead projects the last observed shock forward by ρ^h . For one- and two-period horizons, expected log wages are therefore

$$\begin{aligned} \mathbb{E}_{t-1} \left[\ln w_{ijw_{qt}}(j_i^{MA}, A_i) \right] &= \alpha_{jw,q} + \delta_A^w(A_i) + \delta^{MA} \mathbf{1}\{j^W = j_i^{MA}\} + \rho u_{jw_{q,t-1}}, \\ \mathbb{E}_{t-2} \left[\ln w_{ijw_{qt}}(j_i^{MA}, A_i) \right] &= \alpha_{jw,q} + \delta_A^w(A_i) + \delta^{MA} \mathbf{1}\{j^W = j_i^{MA}\} + \rho^2 u_{jw_{q,t-2}}, \end{aligned}$$

where the first line applies to UK master’s students (one-period horizon) and the second to students at all other destinations (two-period horizon). The expectation collapses to the deterministic component of the wage plus the persistence-discounted current value of the macro residual.

It is worth noting that the belief structure incorporates no forecasts of future immigration policy or job-finding regimes. For the 2010 cohort, it makes master’s destination choice before the March 2011 announcement of the UK policy change, so it cannot anticipate the reform;

upon graduating in June 2011, it can still apply for the PSW visa before the route’s official closure in April 2012. Thus, the 2010 cohort is overall unaffected by the reform in its master’s destination choice. For the 2011 cohort, as noted above, it may or may not adjust the reform given the timing of the announcement. We model this cohort as a mixture of two types, with shares τ and $1 - \tau$: those who adjust their education decisions to the new regime and those who choose as if the pre-reform regime still applied. When they graduate in June 2012, all of them can no longer apply for the PSW visa. For the 2012 and later cohorts, they choose their master’s destination after the announcement and graduate after the PSW route is removed; they are therefore fully aware of and influenced by the new regime and decide their master destination accordingly.

5.4 Econometrics Specification: Selection on Observables

$$\begin{aligned}\mathbb{E}(w_j|MA_j, \mathbf{X}) &= \mathbb{E}(w_j|\mathbf{X}) \\ \mathbb{E}(I_j|MA_j, \mathbf{X}) &= \mathbb{E}(I_j|\mathbf{X})\end{aligned}$$

6 Identification and Estimation

6.1 Identification Sketch

This subsection sketches the sources of variation that identify each component of the model. The identification argument proceeds in three steps, following the temporal logic of the model: wage parameters and beliefs first, then Stage 2 (post-graduation) preferences and stochastic choice set, then Stage 1 (master’s destination) preferences.

Wage parameters and belief formation. The wage equation parameters $(\alpha_{j^W, q}, \delta_A^w, \delta^{MA})$ are identified by realized log wages of working individuals, conditional on the observed choice cell (j^W, q, j^{MA}, A) .²⁴ The location-skill fixed effects $\alpha_{j^W, q}$ are identified by mean log wages within each (j^W, q) cell; the ability shifter δ_A^w is identified by within-cell variation across ability types; and the master’s-match premium δ^{MA} is identified by the wage difference between individuals working in their master’s country and observationally similar individuals working in a different country. The AR(1) persistence parameter ρ and innovation variance σ_e^2 are identified by the time-series autocorrelation of the demeaned wage residual $u_{j^W qt}$ at the country-skill-year level.

²⁴Identification requires that the individual-level wage shock $\nu_{ij^W qt}$ be uncorrelated with the choice shocks ϵ^{MA} and ϵ^W in the structural model.

Given the identified wage parameters, we compute expected log wages $\mathbb{E}_t[\ln w_{j^W_{qt}}]$ for each individual and each Stage-2 alternative. The wage coefficient β^W is then identified from the correlation between expected log wages and Stage-2 choice probabilities across cohorts within each policy regime, holding non-pecuniary utility constant.

Stage 2 structural parameters. The identification of λ^{pre} and λ^{post} is central to the model. The stochastic choice-set probability λ is not directly observed, and it is confounded with the conditional choice probability of the UK high-skilled job: the observed probability factors as

$$\underbrace{\Pr(j^W = (\text{UK}, H) \mid j^{MA} = \text{UK})}_{\equiv P^{(2)}} = \lambda \cdot \frac{\exp(V_2(\text{UK}, H))}{\sum_{j^W \in \mathcal{J}_2(\text{UK})} \exp(V_2(j^W))} + (1 - \lambda) \cdot 0,$$

a mixture of the availability probability λ and the choice probability of the UK high-skilled job, which depends on the amenity η . Choice frequencies alone cannot separate the two.

The resolution is that Stage 1 and Stage 2 respond to the same shock $\Delta\lambda = \lambda^{post} - \lambda^{pre}$ through *different functional forms*, so the relative magnitude of the two responses identifies η . The formal argument is in Appendix F; we sketch the intuition in a two-period version of the model, corresponding to the two policy regimes λ^{pre} and λ^{post} , in which the UK high-skilled job is the only UK employment alternative. Denote by $\pi^*(\eta) = \frac{\exp(V_2(\text{UK}, H))}{\sum_{j^W \in \mathcal{J}_2(\text{UK})} \exp(V_2(j^W))}$ the choice probability when the alternative is in the realized choice set, assumed time-invariant. The observed choice probability in stage 2 is then $P^{(2)} = \lambda \cdot \pi^*(\eta)$, so its change across regimes is

$$P_{post}^{(2)} - P_{pre}^{(2)} = (\lambda^{post} - \lambda^{pre}) \pi^*(\eta).$$

The change in the option value of the UK master's destination is

$$\begin{aligned} IV^{post} - IV^{pre} &= [\lambda^{post} \cdot IV_1^*(\eta) + (1 - \lambda^{post}) \cdot IV_2^*] - [\lambda^{pre} \cdot IV_1^*(\eta) + (1 - \lambda^{pre}) \cdot IV_2^*] \\ &= (\lambda^{post} - \lambda^{pre}) [IV_1^*(\eta) - IV_2^*] \end{aligned}$$

where $IV_1^*(\eta)$ is the option value of UK master when UK job offer appears in the choice set, which depends on the amenity term η . IV_2^* denotes the option value of UK master when UK job offer is not included in the choice set. Thus, the change in the Stage-1 UK master's choice probability satisfies

$$P_{post}^{(1)}(\text{UK}) - P_{pre}^{(1)}(\text{UK}) = \sigma(\varphi(\text{UK}) + \delta \cdot IV^{post}) - \sigma(\varphi(\text{UK}) + \delta \cdot IV^{pre})$$

Here, for simplicity, we assume the outside option is normalized to value 0, but it is

straightforward to extend to other cases. Inverting the logit function,

$$\text{logit}(P_{post}^{(1)}(\text{UK})) - \text{logit}(P_{pre}^{(1)}(\text{UK})) = \underbrace{\delta[IV^{post} - IV^{pre}]}_{\text{option-value change}} = \delta(\lambda^{post} - \lambda^{pre})[IV_1^*(\eta) - IV_2^*].$$

The Stage-2 response is thus linear in $\Delta\lambda$ with slope $\pi^*(\eta)$, while the logit of the Stage-1 response is linear in $\Delta\lambda$ with slope $\delta[IV_1^*(\eta) - IV_2^*]$. Taking the ratio of the two changes nets out $\Delta\lambda$ and leaves an object that depends only on η :

$$\underbrace{\frac{\text{logit}(P_{pre}^{(1)}) - \text{logit}(P_{post}^{(1)})}{P_{pre}^{(2)} - P_{post}^{(2)}}}_{\text{observed}} = \delta \cdot \frac{IV_1^*(\eta) - IV_2^*}{\pi^*(\eta)}.$$

We show that the right-hand side is strictly increasing in η , so η is point-identified. Given η , the observed Stage-2 choice probabilities in the pre- and post-reform regimes recover λ^{pre} and λ^{post} , respectively.

Two caveats apply. First, the argument is based on a simplified model in which only one post-graduation alternative changes (UK high-skilled). In the full model, the UK low-skilled alternative is also removed from the choice set, which affects the option value of the UK master's destination. However, the value of the UK low-skilled alternative is identified from the choice probabilities of the pre-reform cohort, so this change is straightforward to accommodate and the identification argument still holds. Second, the argument assumes $\pi^*(\eta)$ is time-invariant, which may fail in the full model because wages change. Since β^W is identified above, however, changes in $\pi^*(\eta)$ can be incorporated.

The remaining non-wage utilities ξ correspond to alternatives that enter the choice set deterministically; they are identified directly from the shares of students remaining in the host country, working in a third country, returning to India, or continuing education, after adjusting for wage dynamics.

Stage 1 preference parameters. Given the identification of Stage 2 parameters, the inclusive value $IV[j^{MA}; A, O, t]$ can be computed for each individual and each master's destination. The tuition sensitivity parameters (β_1, β_2) are identified by cross-state, cross-year, and cross-destination variation in the rupee-denominated tuition cost $P_{j^{MA}, t} \chi_{j^{MA}, t}$, interacted with origin-state GDP $W_t(O_i)$. The destination-and-type-specific consumption value $\varphi_0[j^{MA}, A]$ is identified by the share of individuals of type A choosing each master's destination j^{MA} , after netting out the predicted contribution of tuition cost and the inclusive value of post-graduation outcomes. With UK as the reference destination, the parameters

(e.g. $\{\varphi_0[US, A]\}$) for each ability type A are identified by the relative shares of US and other countries' master's destinations among individuals of that type, conditional on the model's pecuniary predictions. The discount factor is calibrated as $\delta(j^{MA}) = 0.95^{T(j^{MA})}$, where 0.95 is the annual discount factor and $T(j^{MA})$ is the average length of master's programs at destination j^{MA} in our sample.

6.2 Estimation Procedure

We estimate the model in two steps. The first step is an auxiliary regression that pins down the wage process. The second step is a structural maximum likelihood estimator for the discrete-choice parameters, taking the first-step estimates as given. This subsection describes each step in turn and discusses inference.

6.2.1 Step 1: Wage Process

We estimate the wage equation on the subsample of individuals with observed post-graduation wages, treating the choice cell (j^W, q, j^{MA}, A) as conditioning variables. The estimation proceeds in two sub-steps.

Wage equation. We estimate the parameters of the wage equation by ordinary least squares:

$$\ln w_{ij^{W_{qt}}} = \alpha_{j^W, q} + \sum_{a=2}^{N_A} \delta_A^w(a) \mathbf{1}\{A_i = a\} + \delta^{MA} \mathbf{1}\{j^W = j_i^{MA}\} + e_{ij^{W_{qt}}},$$

where the composite residual $e_{ij^{W_{qt}}} = u_{j^W, qt} + \nu_{ij^{W_{qt}}}$ contains both the macro and individual shocks. The estimation yields parameter estimates $(\hat{\alpha}_{j^W, q}, \hat{\delta}_A^w, \hat{\delta}^{MA})$ that we treat as known in Step 2.

AR(1) process. We then extract the cell-level residual $\hat{u}_{j^W, qt}$ by averaging the OLS residuals within each (j^W, q, t) cell, and estimate the AR(1) persistence and variance parameters by ordinary least squares on the demeaned series:

$$\hat{u}_{j^W, q, t+1} = \rho \hat{u}_{j^W, qt} + \zeta_{j^W, q, t+1}, \quad \zeta \stackrel{\text{iid}}{\sim} \mathcal{N}(0, \sigma_e^2).$$

This yields $\hat{\rho}$ and $\hat{\sigma}_e$. The individual-shock variance $\hat{\sigma}_\nu^2$ is estimated by the residual variance net of the cell-level component.

6.2.2 Step 2: Structural Maximum Likelihood

Given the first-step wage parameters, we estimate the remaining structural parameters θ by maximum likelihood on the joint Stage-1 and Stage-2 choice probabilities. The parameter vector is

$$\theta = (\beta^W, \beta_1, \beta_2, \varphi_0, \xi^{Stay}, \xi^{Third}, \xi^{India,H}, \xi^{PhD}, \xi^{2ndMA}, \eta, \lambda),$$

where φ_0 , ξ^{Stay} , ξ^{Third} , $\xi^{India,H}$, ξ^{PhD} , ξ^{2ndMA} , η , and λ are vectors indexed by the relevant cells—master’s destination, work country, job type, and skill type. Specifically, we assume φ_0 varies with [TODO], and ξ^{Stay} varies with [TODO]. The log-likelihood combines the Stage-1 and Stage-2 conditional logit probabilities:

$$\log L(\theta) = \sum_{i=1}^N \left[\log P_1(j_i^{MA} \mid A_i, O_i, t_i; \theta) + \log P_2(s_{2,i} \mid j_i^{MA}, A_i, t_i; \theta) \right],$$

where, for each individual i with observed master’s destination j_i^{MA} and Stage-2 outcome $s_{2,i}$,

$$\begin{aligned} P_1(j_i^{MA} \mid A_i, O_i, t_i; \theta) &= \frac{\exp(V_1(j_i^{MA}; A_i, O_i, t_i; \theta))}{\sum_{j \in \mathcal{J}_1} \exp(V_1(j; A_i, O_i, t_i; \theta))}, \\ P_2(s_{2,i} \mid j_i^{MA}, A_i, t_i; \theta) &= \frac{\exp(V_2(s_{2,i}; j_i^{MA}, A_i, t_i; \theta))}{\sum_{k \in \mathcal{J}_2(j_i^{MA})} \exp(V_2(k; j_i^{MA}, A_i, t_i; \theta))} \quad \text{for } j_i^{MA} \neq \text{UK}, \\ P_2(s_{2,i} \mid j_i^{MA}, A_i, t_i; \theta) &= \lambda \frac{\exp(V_2(s_{2,i}; \cdot; \theta))}{\sum_{k \in \mathcal{J}_2(\text{UK})} \exp(V_2(k; \cdot; \theta))} + (1 - \lambda) \frac{\mathbb{I}\{s_{2,i} \neq (\text{UK}, H)\} \exp(V_2(s_{2,i}; \cdot; \theta))}{\sum_{k \in \mathcal{J}_2(\text{UK}) \setminus \{(\text{UK}, H)\}} \exp(V_2(k; \cdot; \theta))} \end{aligned}$$

Here $V_2(\cdot)$ and $V_1(\cdot)$ denote the deterministic components of Stage-2 and Stage-1 utility, respectively, as specified in Sections [X] and [X], evaluated at the first-step wage parameters. Stage-1 utility incorporates the inclusive value $IV(j^{MA}; A_i, O_i, t_i)$, the log-sum of the Stage-2 conditional values evaluated at the candidate parameter θ .

6.2.3 Inference

6.2.4 Sample Restrictions

The structural sample is a subset of the full descriptive sample. We conduct the following data restrictions and report the share for each sample selection criteria:

1. Master Destination We first restrict the estimation sample to individuals whose master’s destination is the UK, the US, ANZ (Australia and New Zealand), or Europe; this restriction limits the number of destination-specific parameters to be estimated and preserves statistical power, while retaining the main master’s destinations. It excludes 4,002 individuals (4.2% of the initial sample), reducing the sample from 94,606 to 90,604. Singapore is the largest excluded destination, accounting for 1,919 individuals—approximately 48% of the exclusions.

2. Indian States with too few observations We further exclude students whose bachelor’s degree is from an Indian state with too few observations to support the state-level component of the model: Bihar (68), Tripura (38), Meghalaya (27), Arunachal Pradesh (8), Nagaland (4), Manipur (3), and the Andaman and Nicobar Islands (1). This excludes 149 individuals (0.16% of the 90,604 remaining).

3. Missing post-graduation outcomes We then exclude individuals whose post-graduation outcome is missing, which may reflect students not updating their LinkedIn profiles after completing their education. This restriction excludes 9,347 individuals (10.4% of the remaining sample), reducing it from 90,455 to 81,108. We discuss the implications of these sample restrictions for interpretation in Appendix G.

4. UK low-skilled post-reform We further exclude 203 individuals (0.25% of the sample) out of 81,108 from post-reform cohorts who nonetheless report UK master and low-skilled jobs in the UK, employment not attainable through the post-study or Tier 2 (General) routes the model captures. These observations likely reflect either measurement error, with LinkedIn occupation titles misclassified as low-skilled, or visa routes outside the model, such as permanent residence or spouse visas.²⁵

5. Third country low-skilled Finally, we exclude students who obtain a low-skilled job in a third country—934 individuals (1.15% of the sample), reducing it from 80,905 to 79,971. This restriction implements the model’s abstraction from low-skilled third-country employment: distinguishing skill levels in third countries would increase the dimensionality of the parameter space. Moreover, this margin is both peripheral to the research and doubly thin, third-country employment is itself a minor alternative, and low-skilled jobs account for only 15% of it.

²⁵Figure 1 shows that entry to labor market through these visa routes is negligible.

7 Model Estimate and Model Fit

7.1 Model Estimate

Table 3: Parameter Estimates

<i>Panel A: UK stay amenity $\hat{\eta}_{UK}$ by origin region</i>		
Origin region	$\hat{\eta}_{UK}$ (NG)	$\hat{\eta}_{UK}$ (G)
Southern	-9.350	-0.772
Western	-9.241	-2.096
Northern	-9.228	-0.984
Central	-9.488	0.210
Eastern	-9.308	-1.133
<i>Panel B: UK-Grad availability $\hat{\lambda}_H = \sigma(\hat{\rho}_H)$</i>		
Major group	$\hat{\lambda}_{H,pre}$	$\hat{\lambda}_{H,post}$
STEM	0.334	0.175
NSTEM	0.218	0.116

Table 3 reports the estimates of λ^{pre} and λ^{post} . Under the PSW regime, which guaranteed two years of unconditional stay and required no employer sponsorship, the UK high-skilled alternative enters the realized choice set with probability [xx%] for STEM students and [xx%] for non-STEM students. After the removal of PSW, these probabilities fall to [xx%] and [xx%], declines of [XX] and [XX] percentage points, respectively. The remaining parameter estimates are reported in Appendix H.

We interpret the decline in λ as reflecting three institutional frictions on high-skilled employment introduced unintentionally by the reform. First, the reform shortened the job-search window: the post-graduation grace period fell from 24 months to 4 months. Our data pattern is consistent with this time pressure binding: the share of Indian graduates who stayed in the UK for work and started their first job in the year of master’s completion rose from 48% pre-reform to 64% post-reform.

Second, the reform made employer sponsorship a prerequisite even for graduate-level jobs. A Tier 2 visa requires a Certificate of Sponsorship (CoS) from a licensed employer, and licensing is not free: employers must complete an administrative certification process that small and medium-sized enterprises often find not worthwhile.²⁶ The set of employers able to hire an international graduate therefore shrinks after the abolishment of the PSW visa.

²⁶Migration Advisory Committee (2012) reports consultation evidence that “Partners also told us that some employers had been put off even attempting to bring migrants in because of the constant change to the system and the bureaucracy involved in obtaining CoS, combined with the responsibility of being a sponsor.” Consistent with these fixed costs of sponsorship weighing more heavily on smaller employers, Figure 2 in

Third, sponsorship under Tier 2 raises the cost of hiring a foreign worker relative to the PSW regime, even among licensed employers. A sponsoring employer bears administrative costs that the PSW regime did not impose. It must issue a CoS for each hire and file ongoing compliance reports, notifying the agency of any change affecting a sponsored worker, such as a change in workplace or salary.²⁷ Moreover, sponsorship requires not only that the occupation fall within the RQF level 6 category (graduate-level job) but also that the offer meet the minimum salary threshold (a £20,800 floor under Tier 2 General). These requirements directly raise the per-hire cost of employing an international graduate and operate even for employers already holding a licence.

7.2 Model Fit

Figure 12 plots the share of students choosing a UK master’s across cohorts, in the data (solid line) and as predicted by the model (dashed line). The model reproduces two features of the series. First, it matches the decline in the UK enrollment share that begins with the 2011 cohort. Second, it matches the incomplete drop in that first post-reform cohort (cohort 2011), rather than a full immediate adjustment. The estimated mixing weight τ that governs this transition equals xx%, indicating xx% of individuals have their master’s choice locked into the pre-reform regime, so only the complementary share of students adjusts on impact.

Figure 13 reports the model fit of the distribution of post-graduation outcomes for each master’s destination, separately for the pre-reform regime (cohort 2010 and earlier, left panel) and the post-reform regime (right panel). Blue bars are the shares calculated from the data and orange bars are the shares implied by the model. The top row shows that the model reproduces the post-graduation outcome of the UK branch closely. In particular, it matches the two most salient post-reform shifts: the sharp fall in the share of UK graduates placing into UK high-skilled employment and the offsetting rise in the share returning to India. [TO DO: For the placement share of other countries]

8 Decomposition and Counterfactuals

In section 8.1, we first decompose the reform’s effect on skilled labor supply and education enrollment into three channels. The first is a direct search friction: the policy lowers the job-

Migration Observatory (2024) shows a steep firm-size gradient: the share of workers holding a Skilled Worker visa rises from 1% at micro firms (fewer than 10 employees) and 4% at small firms (10–49 employees) to 15% at large firms (250 or more employees) for year 2016.

²⁷Migration Observatory (2024) reports that small and micro employers voiced concerns that inadvertent non-compliance with monitoring rules could cost them their licences.

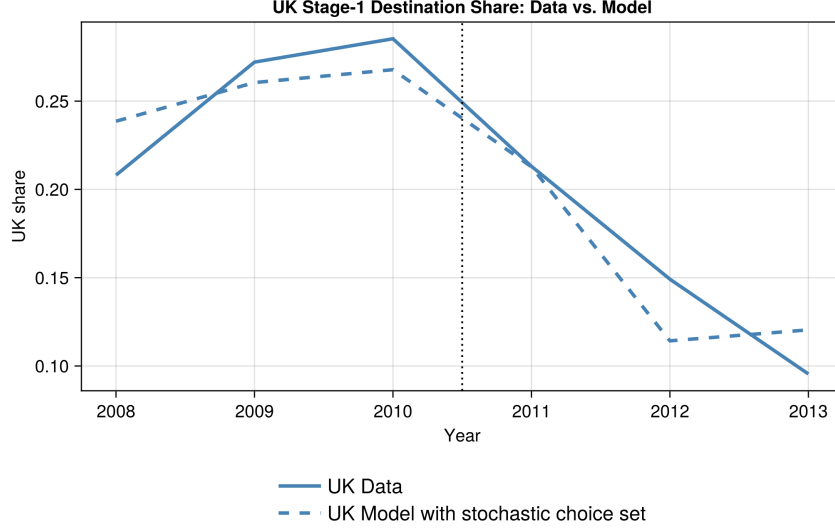


Figure 12: UK Stage 1 Model Fit

finding probability λ , which directly reduces the share of graduates who remain in the UK, holding the enrollment pool fixed. The second is an indirect deterrence channel operating through the education margin. The same fall in λ lowers the option value of a UK master's, so fewer students enroll and the pool shrinks. This channel also shifts the composition of the UK master's pool in both preference and ability. For example, students who still choose the UK for master education post-reform, on average, might relatively value more on UK education itself instead of the opportunity to stay in the UK. The third is the policy's stated mechanism, irrespective of high-skilled arrival rate λ . By curbing low-skilled visa use, the reform lowers the option value of a UK master's directly, independent of its effect on λ . The decomposition that follows quantifies the contribution of each channel to the total decline in both labor and education margins.

In section 8.2, we exploit the structural model and estimates to understand determinants or factors of the immigration visa policy's effect on student enrollment. For example, we try to explain the heterogeneity in the policy response documented in the reduced form. Table 2 shows a larger treatment effect for STEM than for non-STEM students; the model allows us to answer which structural difference between the two groups generates this gap. ²⁸

²⁸For example, one potential reason might be the differential exposure to the sponsorship friction. Although the policy is major-blind, if STEM graduates are more likely to match with small and medium enterprises, which are less likely to hold sponsor licenses, the reform removes a larger share of job vacancies to STEM graduates than non-STEM graduates. Another potential reason is a difference in the baseline job-finding probability λ . With a higher pre-reform λ , STEM students face a larger absolute fall in λ by the same proportional change due to the policy. Hence, they also encounter a larger absolute fall in the option value of a UK master's.

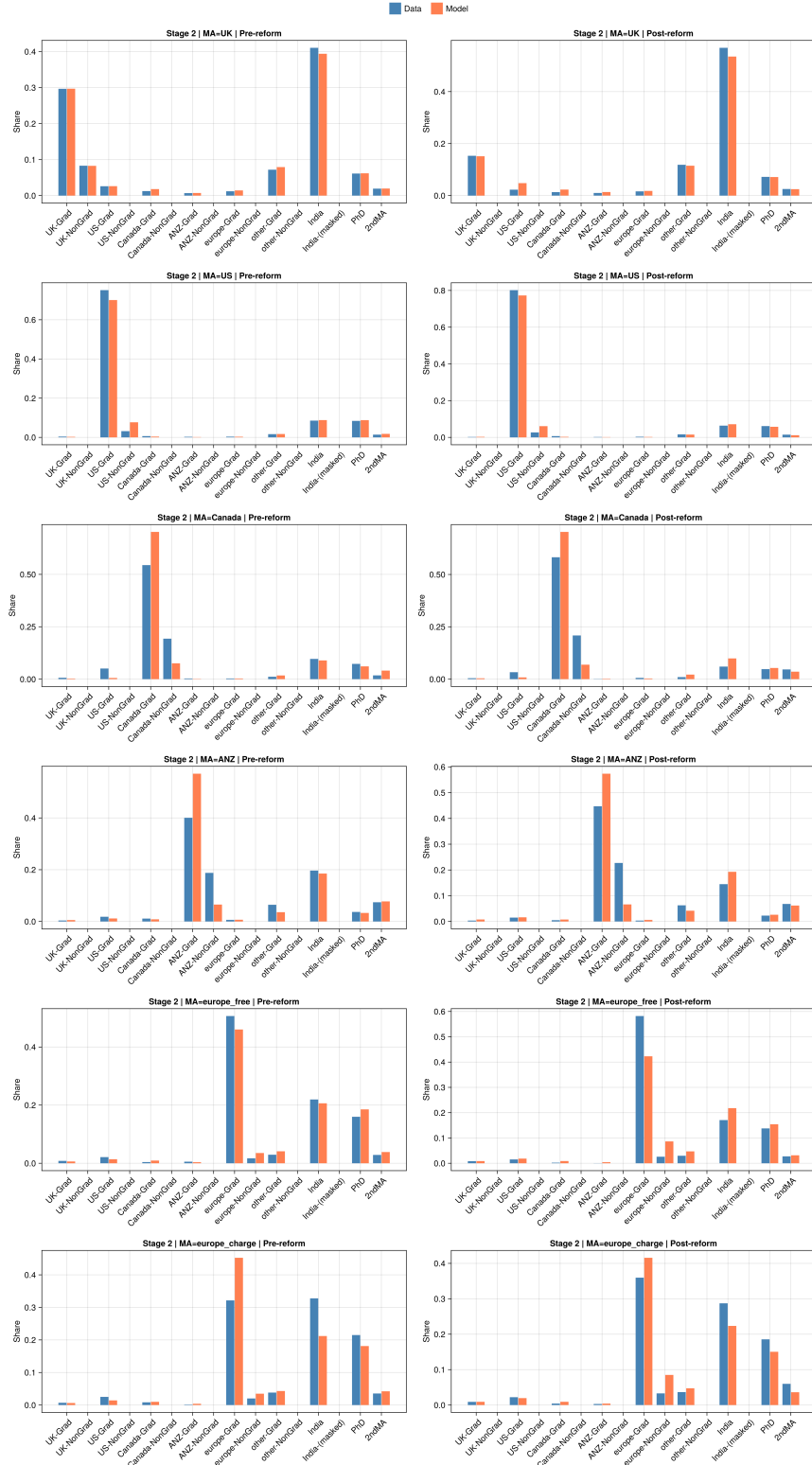


Figure 13: Stage 2 Model Fit by Master Country

Finally, we estimate the trade-off between the inflow of high-skilled immigrants and master’s enrollment. This can be read as: “to reduce foreign skilled labor by X%, how much Indian master’s enrollment (Y %) must the UK forgo?;” Since a host country’s “education exports” depend critically on international students’ post-study employment pathways (Rosenzweig et al., 2006; Bound et al., 2021), this counterfactual explicitly prices the value of local high-skilled labor market access. By quantifying different magnitudes of the negative immigration policy shock on international student enrollment, this analysis provides broader insights into other constrained immigration frameworks like the U.S. H-1B system.

8.1 Decomposition

Figure 14 decomposes the post-reform decline in skilled labor supply into three counterfactuals, each nested in the last. The baseline $S0$ simulates the number of UK high-skilled workers for the post-reform cohorts (2011, 2012, and 2013) under the pre-policy regime (pre-policy λ and the low-skilled job retained in the choice set) and is normalized to 100%.

The first counterfactual holds each individual’s Stage-1 choices fixed and changes only λ^{pre} to λ^{post} , isolating the direct effect of increased hiring friction on placement. This reduces skilled labor supply by 47.5%. The second admits the Stage-1 response, letting individuals anticipate the change in λ but not the removal of the low-skilled option. We split this response into a size effect and a composition effect. For the size effect, we hold the distribution of observables for all UK master’s students at $S0$ and uniformly reduce each cell to match counterfactual enrollment, then recompute the Stage-2 placement distribution. The residual between this and the full counterfactual is the composition effect. The third and the fourth bar in 14 show that the size effect of the λ decline produces a further 25% reduction in skilled labor supply, and the composition effect a further 1%. The same friction also operates on the enrollment margin. Figure 15 shows that anticipation of the increased high-skilled friction alone reduces UK master’s enrollment by 47.4%.

We then allow individuals to also anticipate the removal of the low-skilled host-country option after graduation. The fifth and sixth bars in Figure 14 show that this reduces skilled labor supply only marginally, with a negligible composition effect. The reason lies on its tiny effect on the education margin. Figure 15 shows that removing the low-skilled option lowers UK master’s enrollment only a further 2.1%, from 52.6% to 48.4% of baseline. This follows from the low pre-reform choice probability of the low-skilled job, which reflects its low amenity. Removing that stage 2 option does little to the option value of the UK master’s in the stage 1 choice, and so shifts subsequent skilled labor supply little. The high-skilled friction therefore accounts for 92% (47.4%/51.6%) of the enrollment decline, and the removal

of the low-skilled job for the remaining 8%.

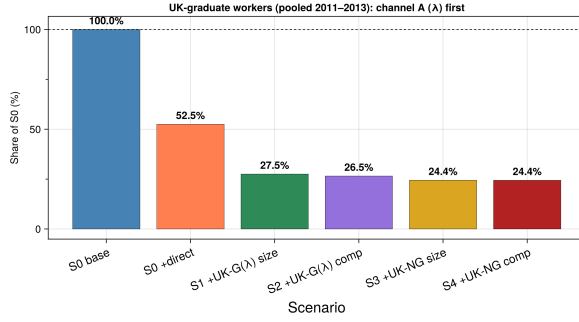


Figure 14: Skilled Worker

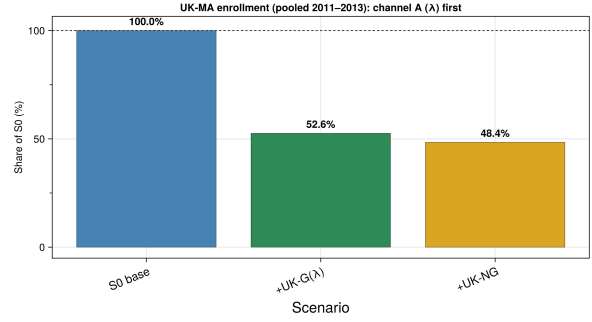


Figure 15: Enrollment

Figures 14 and 15, however, mask important heterogeneity between the STEM and non-STEM groups. Comparing the second bar across Figure 16 and Figure 18, we find that the direct effect is slightly larger for the non-STEM group than for the STEM group (a decline to 54.8% compared to 59.8%). The indirect channel, by contrast, is much larger for the STEM group: it falls by 32.6 percentage points, from 59.8% to 27.2%, compared to a fall from 54.8% to 39.4% for the non-STEM group. This is consistent with the comparison between Figure 17 and Figure 19, where the education margin is much larger for STEM than for non-STEM. We investigate the reason in the next section.

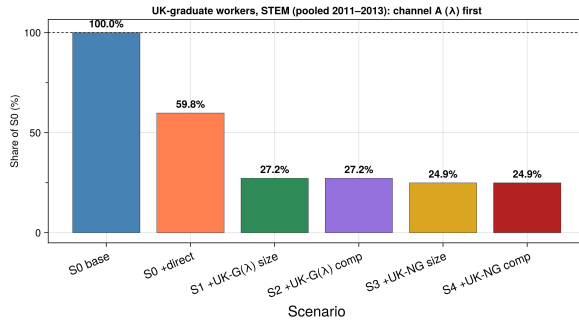


Figure 16: Skilled Worker

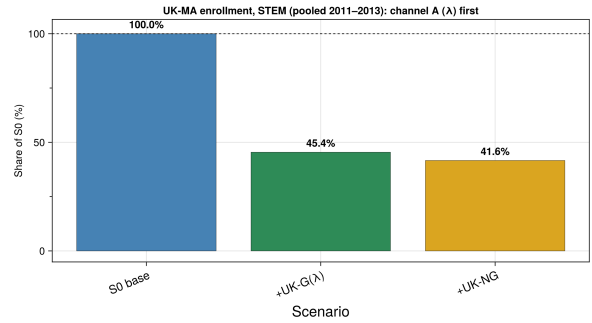


Figure 17: Enrollment

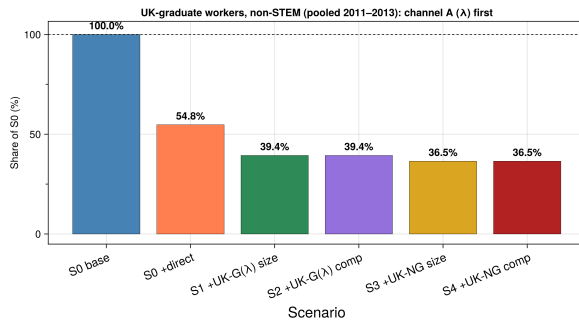


Figure 18: Skilled Worker

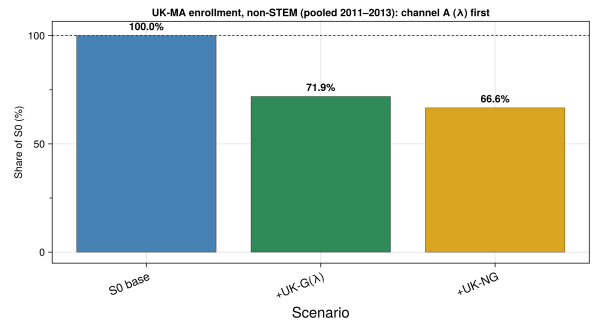


Figure 19: Enrollment

8.2 Counterfactual

The reform reduces λ by a nearly identical proportion for both STEM and non-STEM groups, so the heterogeneous response documented above does not reflect a difference in the size of the shock. Table 3 shows that the proportional decline in λ from the pre- to the post-policy regime is 47.6% for STEM (0.334 to 0.175) and 46.8% for non-STEM, and Figures 17 and 18 show the direct effect is, if anything, slightly larger for non-STEM. This proportional symmetry reflects the major-blind design of the reform.²⁹ Since both groups face a comparable proportional reduction in λ , the STEM/non-STEM difference in outcomes cannot be attributed to the shock itself, but rather to differences in their behavioral patterns and elasticities.

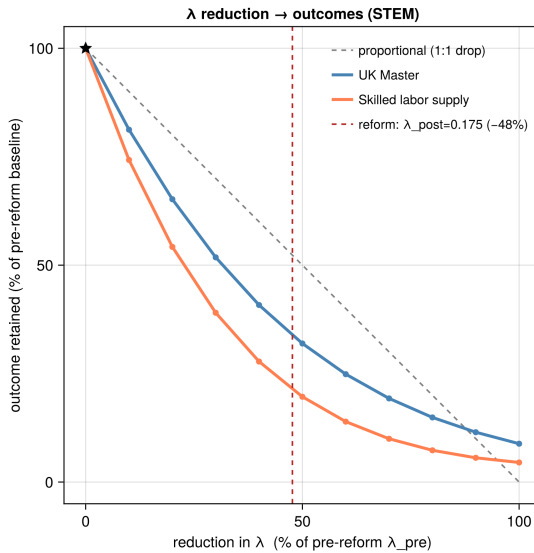


Figure 20: STEM

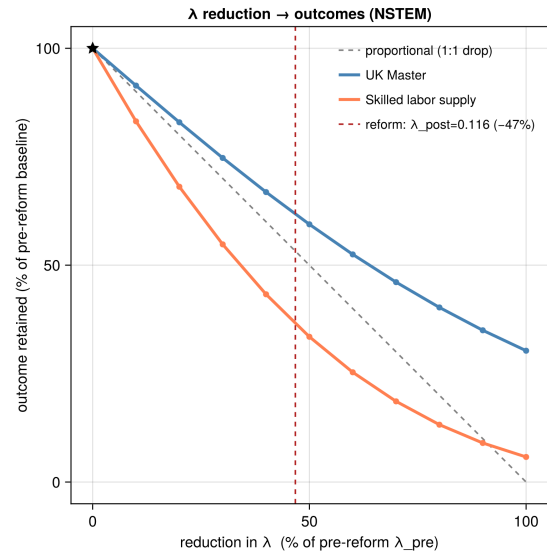


Figure 21: Non-STEM

Figures 20 and 21 trace how UK master's enrollment (blue) and skilled labor supply (orange) respond as the job-finding probability λ is varied exogenously, for STEM and non-STEM graduates respectively. Both axes are normalized to the pre-reform baseline (100%), and the estimated post-reform level ($\sim 52\%$ of pre-reform λ) is marked for reference (red dashed line). We have three takeaways from these graphs.

First, the skilled labor response is more than proportional. The skilled-labor curve lies everywhere below the 45° line, indicating an elasticity greater than one. A given proportional

²⁹Two theoretical arguments support treating the policy as a proportional rather than an absolute change in λ . First, a search-duration model delivers proportionality directly: if offers arrive at rate $\tilde{\lambda}$ over a search window t , then $\lambda \approx \tilde{\lambda}t$, and cutting the window from 24 to 4 months scales λ by the same factor for every group. Second, the theory of labor market tightness does the same. Suppose offers arrive in proportion to the labor market tightness ($\frac{\text{vacancy}}{\text{unemployed}}$), and therefore arrive in proportion to the measure of vacancies open to international candidates. Removing a share of those vacancies (only licensed sponsors willing to bear certification and compliance costs remain) scales λ by that share.

fall in λ produces a larger proportional fall in skilled labor supply. The gap reflects an indirect channel operating through the education margin (the blue curve) where a lower λ also deters enrollment, compounding the direct effect. This education response is stronger for STEM graduates, so the STEM skilled-labor curve bows further below the 45° line than its non-STEM counterpart.

Second, as mentioned above, STEM students have much larger education margin responses (indicated by the steeper blue curve) to λ than non-STEM students. The first important factor contributing to these heterogeneous slopes is the pre-reform level of λ . Figure 22 runs a thought experiment along a single dimension of λ^{pre} . Suppose we hold all other parameters fixed at their STEM values and implement a common 52% cut in λ , how the UK master enrollment varies only with the pre-reform availability λ^{pre} . The result shows that the enrollment response first increases with λ^{pre} . This is because the same **proportion** cut in λ^{pre} destroys a larger **absolute** value of UK employment opportunity when availability λ starts high. Thus, the option value of a UK degree falls more and makes the enrollment response grow with the pre-reform level λ^{pre} . Both STEM and non-STEM begin with relatively low λ^{pre} and so sit on the increasing region of this curve, but STEM's pre-reform λ is closer to the peak, leading to larger enrollment response for STEM. The relationship eventually turns over. When λ^{pre} is high enough, the incremental reduction in option value is offset by the diminishing marginal effect of λ at the extreme, and the response decreases with λ^{pre} at the upper end.

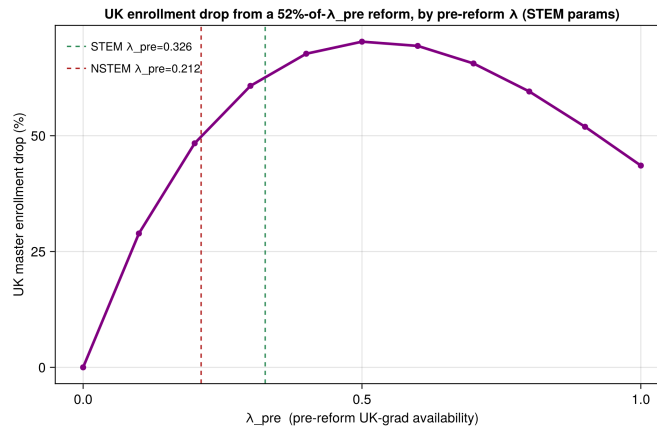


Figure 22: Original Lambda

1

The vertical red dashed line in Figure 22 shows that, holding all other parameters at their STEM values and feeding in only the non-STEM λ^{pre} , the model predicts an enrollment loss of 50%, larger than the actual 38% loss for non-STEM shown in Figure 19. Changing the

pre-reform level of λ alone thus still *over*-predicts the non-STEM response. The remaining gap reflects the different consumption value of a UK education between STEM and non-STEM students, which is higher for non-STEM. For example, the estimated education taste for the US master's relative to the UK master's is 4.7, 4.8, and 5.3 for Tier-1, Tier-2, and Tier-3 STEM students, much higher than the corresponding 3.2, 2.7, and 2.4 for non-STEM. The higher relative value placed on the UK master's by non-STEM students raises their pre-reform UK master's share (a larger denominator), so for the same absolute change in inclusive value the proportional enrollment response is smaller for non-STEM. In sum, a group or destination has higher elasticity to the policy shock when (i) its pre-reform stay rate is high, so the policy erodes a larger absolute share of option value, and (ii) its pre-reform baseline share in the international education market is low, producing a larger proportional response.

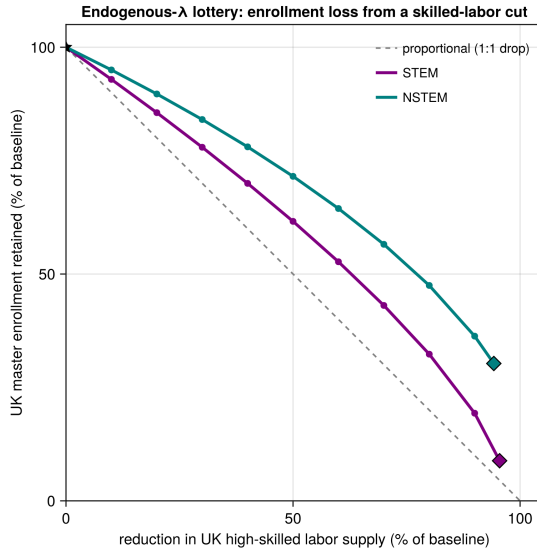


Figure 23: Elasticity

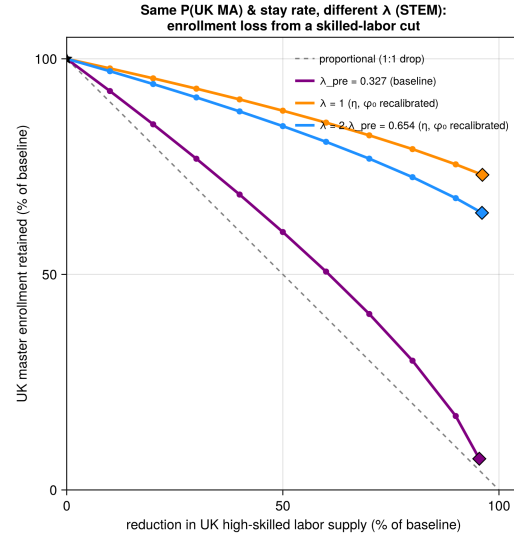


Figure 24: Elasticity (Observationally Equivalent)

Then we illustrate the trade-off between skilled labor supply and master's enrollment. Suppose the government seeks to reduce high-skilled immigration by lowering the job-finding probability λ (e.g., through an H-1B-style lottery on offers). Doing so inevitably depresses student enrollment, and hence the competitiveness of the UK's education export sector. Figure 23 plots illustrate this “price” to the employment opportunity shrinkage. Unlike Figures 20 and 21, where λ is varied directly as an exogenous parameter, here the government controls the realized quantity of skilled labor it will admit (x-axis), and we solve backward for the λ and enrollment (y-axis) consistent with it. The simulation indicates that if the UK cuts the offers available to international students by 20%, enrollment falls by 14% for STEM students and 10% for non-STEM students. A larger cut of 50% reduces enrollment by 38%

for STEM and 27% for non-STEM.

Moreover, this result cannot be pinned down by the cross-sectional shares. To show this, Figure 24 traces a family of parameter setups that induce observationally equivalent baseline outcomes to the STEM group. Specifically, we vary λ^{pre} while adjusting the consumption value φ and the amenity η to hold the observed Stage-1 and Stage-2 probabilities fixed. Raising λ , for instance, requires lowering the amenity of UK high-skilled jobs η to preserve the Stage-2 choice probability as well as lowering the relative education consumption value of the UK master's φ to preserve the Stage-1 choice probability. For each setup, we recompute the trade-off between skilled labor supply and master's enrollment. Although these setups fit the data identically, they imply very different enrollment sensitivities. When the stay option is a rare, high-prize lottery (low λ , high η), enrollment responds sharply to its availability, whereas when it is a common, low-prize option (high λ , low η), it responds far less. The next paragraph explains why.

Consider the stage-1 decision as integrating the option value over unrealized stage-2 taste draws ϵ_2 , in particular the stage-2 taste of UK high-skilled job $\epsilon_2(\text{UK,H})$. A cut in λ destroys the UK high-skilled outcome only in the $\epsilon_2(\text{UK,H})$ states where the student would have accepted the offer, so the option value loss is the mass of destroyed acceptance states times the average utility gain those acceptance states carried. The observational equivalence of the Stage 2 choice probability condition pins down the first term. This is because the equivalent stay rate $\lambda \cdot P(\text{UK,H})$ suggests that a proportional cut in λ destroys the same mass of acceptance states in every regime. The regimes differ only in the second term. In the rare, high-prize regime, every destroyed state of $\epsilon_2(\text{UK,H})$ carries a very high value in high-skilled offer. Yet, in the common, low-prize regime, acceptance occurs only at favorable taste draws, and even in those states, the UK high-skilled option only sits marginally above the outside alternatives. Thus, each destroyed acceptance state by declining λ takes away little value, whereas in the rare, high-prize regime every destroyed state carries high value. The same proportional cut in λ therefore removes greater option value, and induces a correspondingly larger enrollment response in the rare, high-prize regime. To sum up, the Stage-2 choice shares pin down the mass of destroyed acceptance states and therefore the direct effect of a potential policy, but they do not tell the value those states carried. In other words, observing the cross-sectional shares alone are not sufficient to recover the value of an employment opportunity. Recovering the price of shrinking labor-market access on enrollment requires other variation (e.g. reform-based) that separately identifies job finding probability λ and amenity η .

Table 4: Counterfactual

	Counterfactual lever			
	UK wage $\uparrow$	$\lambda_e \uparrow$	$\lambda_r \uparrow$	UK tuition $\downarrow$
<i>Panel A. Shock magnitude: 10 percent</i>				
Δ UK master's enrollment	7.53%	1.03%	13.26%	7.15%
Δ UK tuition income (£m/yr)	12.1	1.7	21.3	-5.7
Δ UK elite labor supply	7.17%	11.20%	12.08%	7.01%
Δ UK regular labor supply	8.10%	0.89%	25.82%	7.39%
Δ UK high-skill workers (persons/yr)	224	48	691	206
Δ UK GDP (£bn)	0.208	0.037	0.508	0.194
<i>Panel B. Shock magnitude: 50 percent</i>				
Δ UK master's enrollment	38.31%	5.23%	75.66%	38.36%
Δ UK tuition income (£m/yr)	61.5	8.4	121.5	-49.4
Δ UK elite labor supply	36.40%	58.28%	67.70%	37.51%
Δ UK regular labor supply	41.30%	4.48%	174.11%	39.96%
Δ UK high-skill workers (persons/yr)	1,143	246	4,629	1,111
Δ UK GDP (£bn)	1.060	0.187	3.223	1.046

9 Conclusion

This paper has examined the United Kingdom's 2012 abolition of the Post-Study Work visa, a reform that removes international students' two years of unconditional stay post-graduation, aiming to eliminate their low-skilled employment pathways in the U.K.. However, our analysis indicates that this policy substantially raised hiring frictions for graduate-level employment as well. Combining a novel longitudinal dataset of 94,606 Indian master's students with a two-stage structural model of education and labor-market decisions, we estimate that the reform reduced UK graduate-level employment of Indian students by 70%, together with master's enrollment by 53%, equivalent to roughly £110 million in foregone annual tuition revenue. Through decomposing the aggregate decline in UK graduate-level employment, we find that the direct effect of higher post-graduation search frictions on graduate-level UK employment accounts for 63% and the indirect deterrence on the education choices operating through anticipated graduate-level frictions accounts for an additional 34%. Counterfactual simulations quantify the price of shrinking employment opportunities. Each 1% loss of skilled positions comes with a 0.7% enrollment loss among STEM students and a 0.5% loss among non-STEM students; baseline differences in labor-market access account for most of this heterogeneity.

Our findings carry broader implications for the design of high-skilled immigration pol-

icy. Reforms that target what appear to be low-skilled visa channels may generate sizable unintended distortions in high-skilled matching and retention that counteracts stated policy objectives. Flexible post-study work routes function not only as pathways to low-skilled employment but also as job-search windows for graduate-level employment. Moreover, it can serve as low-compliance hiring channels through which employers absorb international graduates without bearing the administrative costs of sponsorship and certification. This institutional interplay is not specific to the United Kingdom; the structural trade-offs identified here directly map into key debates surrounding high-skilled labor supply in other host countries, most notably the operational interactions between Optional Practical Training (OPT) regulations and the H-1B visa lottery in the United States. In addition, our findings offer broader methodological implications for dynamic discrete-choice models in which future options become available only **stochastically**. Without additional exclusion restrictions, the data cannot pin down the trade-off between the upstream benefit and the downstream cost (or vice versa). The same baseline choice probabilities can be rationalized by multiple parameter families, each implying a very different trade-off elasticity depending on how much of the observed behavior is attributed to **constraints** compared to **preferences**. This insight is relevant across a wide range of empirical settings in which probabilistic restrictions on downstream choice sets shape upstream investment decisions, such as entrepreneurial entry under acquisition policy.

Several questions remain for future work. Our analysis focuses on the first employment spell after master’s completion, leaving the medium- and long-run trajectories of international graduates open for further investigation. If low-skilled host-country employment serves as a stepping stone into skilled occupations over time, then closing the low-skilled entry route may depress long-run skilled retention by more than our model captures; conversely, if post-reform graduates entering skilled jobs later move into lower-skilled ones, the reform may have induced occupational mismatch by channeling marginal applicants into ill-fitting skilled positions. Quantifying these lifecycle dynamics is a valuable direction for understanding immigration policy and the global competition for talent.

References

- Abarcar, P. and C. Theoharides (2024). Medical worker migration and origin-country human capital: Evidence from us visa policy. *The Review of Economics and Statistics* 106(1), 20–35.
- Adda, J., C. Dustmann, and J.-S. Görlach (2022). The dynamics of return migration, human capital accumulation, and wage assimilation. *The Review of Economic Studies* 89(6), 2841–2871.
- Altonji, J. and Z. Zhu (2025). Returns to specific graduate degrees: Estimates using texas administrative records. Technical report, National Bureau of Economic Research.
- Altonji, J. G., P. Arcidiacono, and A. Maurel (2016). The analysis of field choice in college and graduate school: Determinants and wage effects. In *Handbook of the Economics of Education*, Volume 5, pp. 305–396. Elsevier.
- Altonji, J. G. and L. Zhong (2021). The labor market returns to advanced degrees. *Journal of Labor Economics* 39(2), 303–360.
- Amanzadeh, N., A. Kermani, and T. McQuade (2024). Return migration and human capital flows. Technical report, National Bureau of Economic Research.
- Amuedo-Dorantes, C. and D. Furtado (2019). Settling for academia? h-1b visas and the career choices of international students in the united states. *The Journal of Human Resources* 54(2), 401–429.
- Amuedo-Dorantes, C., D. Furtado, and H. Xu (2019). Opt policy changes and foreign born stem talent in the us. *Labour Economics* 61, 101752.
- Amuedo-Dorantes, C., K. Shih, and H. Xu (2023). The implications of optional practical training reforms on international student enrollments and quality. *Economic Inquiry* 61(2), 253–281.
- Arcidiacono, P., E. Aucejo, A. Maurel, and T. Ransom (2025). College attrition and the dynamics of information revelation. *Journal of Political Economy* 133(1), 53–110.
- Arenas, A. (2021). Human capital portability and international student migration. *Journal of Economic Geography* 21(2), 195–229.
- Barseghyan, L., F. Molinari, and M. Thirkettle (2021). Discrete choice under risk with limited consideration. *American Economic Review* 111(6), 1972–2006.
- Batista, C., D. M. Costa, P. Freitas, G. Lima, and A. B. Reis (2025). What matters for the decision to study abroad? a lab-in-the-field experiment in cape verde. *Journal of Development Economics* 173, 103401.
- Beine, M., M. Delogu, and L. Ragot (2020). The role of fees in foreign education: evidence from italy. *Journal of Economic Geography* 20(2), 571–600.
- Beine, M., R. Noël, and L. Ragot (2014). Determinants of the international mobility of students. *Economics of Education review* 41, 40–54.
- Beine, M., G. Peri, and M. Raux (2023). International college students’ impact on the us skilled labor supply. *Journal of Public Economics* 223, 104917.
- Beine, M., G. Peri, and M. Raux (2024). The contribution of foreign master’s students to us start-ups. Technical report, National Bureau of Economic Research.
- Bernstein, S., R. Diamond, A. Jiranaphawiboon, T. McQuade, and B. Pousada (2022). The contribution of high-skilled immigrants to innovation in the united states. Technical report, National Bureau of Economic Research.

- Bontemps, C., J.-M. Robin, and G. J. Van den Berg (2000). Equilibrium search with continuous productivity dispersion: Theory and nonparametric estimation. *International Economic Review* 41(2), 305–358.
- Bound, J., B. Braga, J. M. Golden, and G. Khanna (2015). Recruitment of foreigners in the market for computer scientists in the united states. *Journal of labor economics* 33(S1), S187–S223.
- Bound, J., B. Braga, G. Khanna, and S. Turner (2020). A passage to america: University funding and international students. *American Economic Journal: Economic Policy* 12(1), 97–126.
- Bound, J., B. Braga, G. Khanna, and S. Turner (2021). The globalization of postsecondary education: The role of international students in the us higher education system. *Journal of Economic Perspectives* 35(1), 163–184.
- Bound, J., M. Demirci, G. Khanna, and S. Turner (2015). Finishing degrees and finding jobs: Us higher education and the flow of foreign it workers. *Innovation Policy and the Economy* 15(1), 27–72.
- Brinatti, A. and X. Guo (2025). Third-country effects of us immigration policy. Staff Working Paper 2023-60, Bank of Canada. Last updated November 6, 2024.
- Chen, M. (2024). The value of us college education in global labor markets: Experimental evidence from china. *Management Science* 70(2), 1276–1300.
- Chen, M., J. Howell, and J. Smith (2023). Best and brightest? the impact of student visa restrictiveness on who attends college in the us. *Labour Economics* 84, 102385.
- Clemens, M. A., J. Neufeld, and A. M. Nice (2025). Brain freeze: How international student exclusion will shape the stem workforce and economic growth in the united states.
- Crown, D., A. Faggian, and J. Corcoran (2020). Foreign-born graduates and innovation: Evidence from an australian skilled visa program. *Research Policy* 49(5), 103945.
- Demirci, M. (2019). Transition of international science, technology, engineering, and mathematics students to the us labor market: The role of visa policy. *Economic Inquiry* 57(3), 1367–1391.
- Demirci, M. (2020). International students and labour market outcomes of us-born workers. *Canadian Journal of Economics/Revue canadienne d'économique* 53(4), 1495–1522.
- Dimmock, S. G., J. Huang, and S. J. Weisbenner (2022). Give me your tired, your poor, your high-skilled labor: H-1b lottery outcomes and entrepreneurial success. *Management Science* 68(9), 6950–6970.
- Dustmann, C., A. Lindner, U. Schönberg, M. Umkehrer, and P. Vom Berge (2022). Reallocation effects of the minimum wage. *The Quarterly Journal of Economics* 137(1), 267–328.
- Flinn, C. and J. Heckman (1982). New methods for analyzing structural models of labor force dynamics. *Journal of econometrics* 18(1), 115–168.
- Flynn, R., B. Glennon, R. Murciano-Goroff, and J. Xiao (2024). Building a wall around science: The effect of us-china tensions on international scientific research. Technical report, National Bureau of Economic Research.
- French, E. and C. Taber (2011). Identification of models of the labor market. In *Handbook of labor economics*, Volume 4, pp. 537–617. Elsevier.
- Ganguli, I. and M. MacGarvie (2025). International students, immigration policies and implications for innovation. Working Paper 34212, National Bureau of Economic Research.

- Glennon, B. (2024). How do restrictions on high-skilled immigration affect offshoring? evidence from the h-1b program. *Management Science* 70(2), 907–930.
- Heckman, J. (1990). Varieties of selection bias. *The American Economic Review* 80(2), 313–318.
- Hu, S. and J. V. Winters (2025). Growing from the stem? opt classification and international students in economics. Discussion Paper 18336, IZA Institute of Labor Economics.
- Hunt, J. (2011). Which immigrants are most innovative and entrepreneurial? distinctions by entry visa. *Journal of Labor Economics* 29(3), 417–457.
- Jia, R., G. Khanna, H. Li, and Y. Xu (2025). The ripple effects of china’s college expansion on american universities. Working Paper 34391, National Bureau of Economic Research.
- Kato, T. and C. Sparber (2013). Quotas and quality: The effect of h-1b visa restrictions on the pool of prospective undergraduate students from abroad. *Review of Economics and Statistics* 95(1), 109–126.
- Kerr, S. P., W. R. Kerr, and W. F. Lincoln (2015). Skilled immigration and the employment structures of us firms. *Journal of Labor Economics* 33(S1), S147–S186.
- Kerr, W. R. and W. F. Lincoln (2010). The supply side of innovation: H-1b visa reforms and us ethnic invention. *Journal of Labor Economics* 28(3), 473–508.
- Khanna, G. (2025). From asia, with skills. Working Paper 34449, National Bureau of Economic Research.
- Khanna, G. and N. Morales (2025). The it boom and other unintended consequences of chasing the american dream. Technical Report 460, Center for Global Development. Updated January 2025.
- Khanna, G., K. Shih, A. Weinberger, M. Xu, and M. Yu (2025). Trade liberalization and chinese students in us higher education. *Review of Economics and Statistics* 107(5), 1291–1309.
- Manzini, P. and M. Mariotti (2014). Stochastic choice and consideration sets. *Econometrica* 82(3), 1153–1176.
- Migration Advisory Committee (2012, February). Limits on tier 2 (general) for 2012/13 and associated policies. Report, Home Office, United Kingdom. Accessed: 2026-07-07.
- Migration Observatory (2024, January). Licenced visa sponsors. Commentary, The Migration Observatory. Accessed 7 July 2026.
- Peri, G., K. Shih, and C. Sparber (2015a). Foreign and native skilled workers: What can we learn from the h-1b visa lotteries?
- Peri, G., K. Shih, and C. Sparber (2015b). Stem workers, h-1b visas, and productivity in us cities. *Journal of Labor Economics* 33(S1), S225–S255.
- Rosenzweig, M. R., D. A. Irwin, and J. G. Williamson (2006). Global wage differences and international student flows [with comments and discussion]. In *Brookings trade forum*, pp. 57–96. JSTOR.
- Ruiz, N. G. and A. Budiman (2018). Number of foreign college students staying and working in us after graduation surges. *Pew Research Center*.
- Shih, K. (2016). Labor market openness, h-1b visa policy, and the scale of international student enrollment in the united states. *Economic Inquiry* 54(1), 121–138.
- Shih, K. (2017). Do international students crowd-out or cross-subsidize americans in higher education? *Journal of Public Economics* 156, 170–184.
- Stuen, E. T., A. M. Mobarak, and K. E. Maskus (2012). Skilled immigration and innova-

- tion: evidence from enrolment fluctuations in us doctoral programmes. *The Economic Journal* 122(565), 1143–1176.
- Zhu, J. L. (2024). Comparative immigration policies and the resource effects of international students in us higher education. Working Paper.

Tables

Figures

A Appendix A: LinkedIn Data

Figure 28 plots the total number of Indian students enrolled in U.S. master’s programs constructed from our Revelio Labs LinkedIn profile dataset. For comparison, Figure 29, originated from Figure 2 (b) in Amuedo-Dorantes et al. (2023), presents average international program enrollment by degree level and field of study, utilizing administrative records from the Student and Exchange Visitor Information System (SEVIS). The time-series variation in our sample closely mirrors the aggregate administrative trends depicted in Figure 29. Specifically, both datasets reveal that STEM enrollment remained relatively flat prior to 2012, underwent a sharp expansion starting in 2013, and experienced a mild contraction in 2015. In contrast, non-STEM enrollment maintains a persistently lower and stable trajectory across the sample period in both sources. This tight alignment not only validates the representativeness of our LinkedIn-derived sample, but also suggests the great share of Indian students in U.S. STEM master’s enrollment.

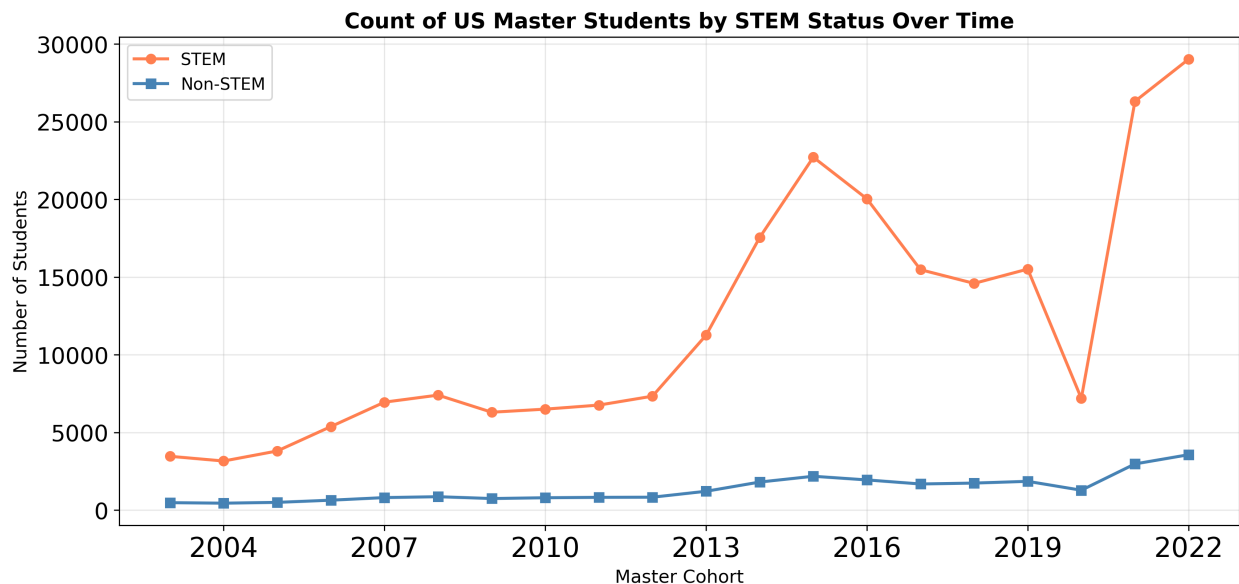
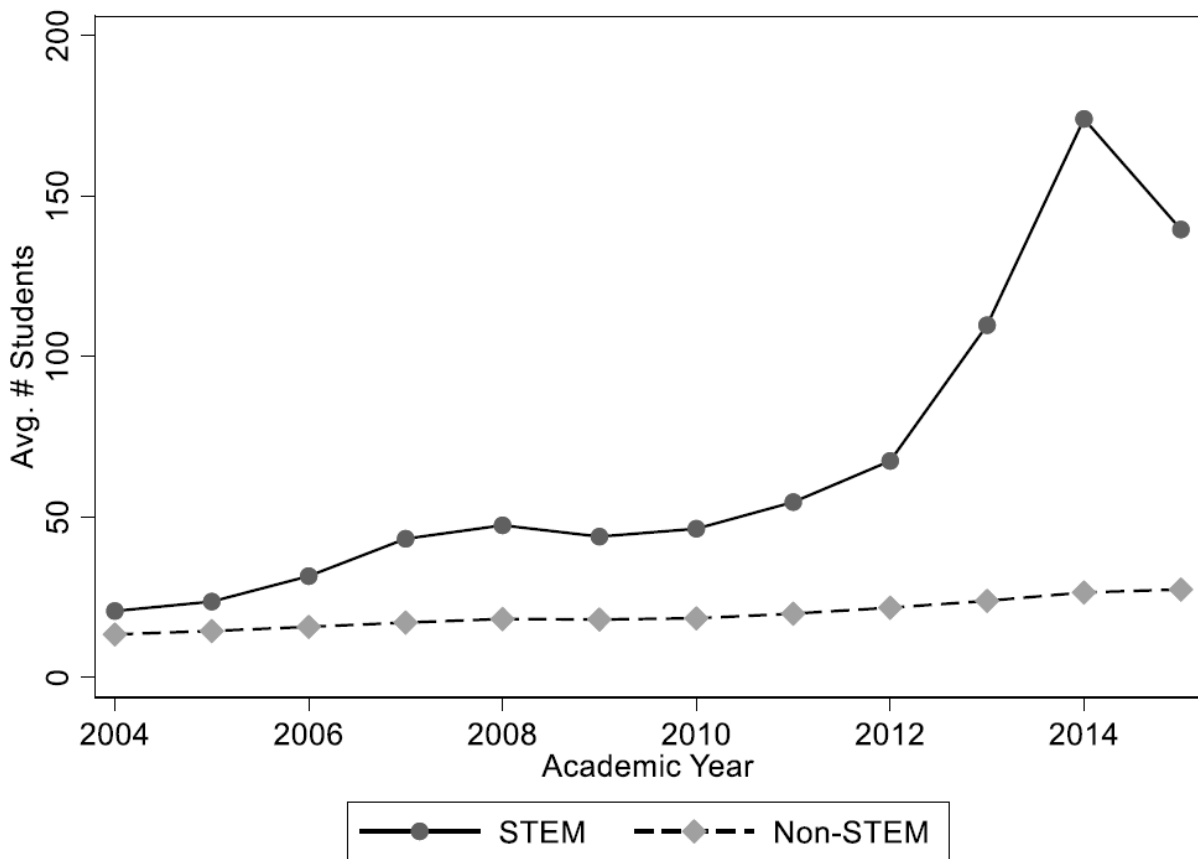


Figure 25: US Master Enrollment



(b) Master's

Figure 26: US Average Enrollment (Figure 2 (b) from Amuedo-Dorantes et al, 2023)

B Appendix B: Indian Bachelor Ranking Data

B.1 B.1 Institutional Structure and Admissions in Indian Higher Education

Indian higher education institutions can be classified into four legal categories based on their founding authority and statutory status: Central Universities, State Universities, Private Universities, and Deemed to be Universities. Central Universities are established by the central government, State Universities by state governments, and Private Universities by private entities under state legislation, while Deemed to be Universities are institutions granted university status through statutory authorization. Notably, these legal categories do not directly correspond to a quality hierarchy, and substantial heterogeneity exists within each category.

A distinct statutory designation within the higher education system is that of Institutions of National Importance (INIs), which are established through parliamentary legislation. INIs include selected public professional institutions, primarily in engineering, medicine, and management, and are consistently ranked among the top institutions in their respective fields. They operate under separate legal frameworks and receive direct funding from the central government, with relatively centralized governance structures. In constructing our institutional quality measures, INI status is explicitly incorporated into the imputation procedure.

Undergraduate admissions in India are organized primarily by field of study. Major professional disciplines, including engineering, medicine, law, and management, rely on separate national or multi-state entrance examination systems. Candidates take examinations corresponding to their chosen field and participate in centralized seat allocation based on national rank and submitted preferences. The admission process is administered through designated examination authorities and centralized allocation mechanisms. As a result, admission systems operate independently across fields, and no single entrance examination covers all institutions and programs.

The combination of multiple institutional categories and discipline-specific admission systems implies the absence of a unified quality evaluation framework. Admission procedures and information disclosure practices vary across institution types and fields, leading to fragmented observable data. Some institutions appear in national ranking systems, others are observable only through field-specific examination outcomes, and some lack publicly reported ranking or admission information. Measuring institutional quality therefore requires integrating information from multiple sources. The next subsection describes the two sources of quality measures used in this study.

B.2 B.2 Quality Measures

Evaluating institutional quality in the Indian higher education system is complicated by discipline-specific admissions mechanisms and incomplete coverage of ranking data. To construct consistent measures of institutional quality, we rely on two complementary sources: (i) official national rankings, and (ii) discipline-specific entrance examination cutoffs.

B.2.1 B.2.1 National Institutional Ranking Framework (NIRF)

The National Institutional Ranking Framework (NIRF) was launched in 2015 by the Ministry of Education (MOE) as a standardized framework for ranking higher education institutions in India. It provides an annual ranking based on a uniform evaluation methodology.

NIRF publishes rankings across multiple categories, including an overall category as well as discipline-based categories such as engineering, law, management, and medicine. Institutions are evaluated and ranked within their respective categories. The ranking is determined by a composite score constructed from five dimensions: Teaching, Learning and Resources; Research and Professional Practice; Graduation Outcomes; Outreach and Inclusivity; and Perception. Each dimension aggregates multiple quantitative indicators, and the final score is calculated as a weighted average of these components.

Participation in NIRF requires institutional data submission, and published rankings typically include only a limited number of institutions within each category. In our construction of institutional quality measures, we implement a score imputation procedure to combine information across categories. However, due to the limited coverage of NIRF, a substantial number of institutions do not receive an observed ranking score.

B.2.2 B.2.2 Entrance Examination Cutoffs

Admission to undergraduate engineering and law programs in India is conducted through centralized national entrance examinations followed by centralized seat allocation procedures.

In engineering, admissions are based on the Joint Entrance Examination (JEE). The examination is administered by the National Testing Agency (NTA), which conducts the test, evaluates performance, and publishes standardized All-India Rank lists. Seat allocation is conducted separately through a centralized counseling mechanism under which candidates submit ordered program preferences and seats are assigned based on rank and availability. Program-level opening and closing ranks are publicly reported for each institution and admission round.

In law, admissions to National Law Universities are conducted through the Common Law Admission Test (CLAT). The examination generates a national rank list, and seat allocation

is conducted centrally among participating institutions. Opening and closing ranks are publicly reported for each institution and admission round.

We use the publicly reported opening and closing ranks from these examinations to construct rankings for institutions that admit students through these examination systems, which serve as a supplementary measure of institutional quality.

- C Appendix C: Occupation List
- D Appendix D: Quasi-Experimental Specification
- E Appendix E: Tuition Construction
- F Appendix F: Identification Proof
- G Appendix G: Sample Selection
- H Appendix H: Estimates

I Appendix I: Administrative Statistics for Indian Students in the U.K.

In this appendix, we exploit 12 auxiliary administrative UK visa and higher education datasets to characterize the full population of Indian students in the U.K. before and after the policy. These sources provide important insight into students at low-quality institutions and into those entering low-skilled employment, neither of whom we observe in the LinkedIn panel. These data buy us two valuable pieces of information. First, because curbing low-skilled stay was a stated objective of the 2012 PSW abolition, they allow us to assess whether the reform achieved that main objective. Second, once low-skilled entrants are netted out, the administrative counts provide a denominator against which to measure the coverage and representativeness of the LinkedIn sample within the high-skilled population.

Specifically, we use two features of these data. First, visa record data allow us to trace the visa-status path (in an aggregate manner) for each study-entry cohort, defined as all Indian nationals first entering the United Kingdom on a Tier 4 study visa in a given calendar year. Second, the wedge between the visa-application counts and the higher education enrollment counts helps us identify the share of Indian students enrolled in each type of institution. In the following parts, Section I.1 describes the three most important datasets and how we use each dataset to construct an accounting model. Section I.2 presents the results and their implications. Section I.3 discusses what these results imply for the coverage of our main LinkedIn data.

I.1 Data Source

Among the twelve administrative datasets we use, the most important is *Migrant Journey*, which tracks the pathways individuals take through the UK immigration system from initial entry on a visa to either departure or the grant of indefinite leave to remain (ILR). The dataset records the end-of-calendar-year status of each calendar-year cohort first entering the United Kingdom on a study or work visa. For each study-visa cohort, we observe the total number of entrants in a given calendar year, and the number of transitions from a student visa to each other visa status at the end of year one and at the end of year two. These counts are reported by nationality, so we obtain figures specific to Indian students. In Figure 27, these counts pin down the total under “ENTRY (Ref)” and the size of each block under “end of year 1” and “end of year 2”.

Our second source is UK study-visa application data from the Confirmation of Acceptance for Studies (CAS) system, which reports quarterly counts of visa applications by nationality,

application type (new visa or extension), and sponsor type (higher education institution or college). Two of these dimensions are particularly important for us. Most importantly, the sponsor type dimension lets us isolate Indian postgraduate students on taught master’s programs at recognized higher education institutions (HEI), separating them from entrants to other college sectors (e.g., further education and language schools) and franchised colleges (See next paragraph). This appears as the sum of the light purple brown block in Figure 27³⁰. Even within HEI-sponsored visa applications, not every application corresponds to a taught-master entrant, representing a small share for courses below degree level (e.g., pre-session English and foundation programs), and a small share of granted visas never used for entry (forfeited). We net both out and display them as the teal and grey blocks in Figure 27. The two shares are set with auxiliary data described below. In addition, the application-type dimension identifies the share of taught master’s entrants arriving on in-country extensions rather than new visas. This distinction is necessary for reconciling the two sources. *Migrant Journey* cohorts consist of first entries only, so extension students, the second block in the first line of Figure 27, fall outside of *Migrant Journey* entry and should not be counted.

The third source is the UK Higher Education Statistics Agency (HESA), which reports the stock of enrolled students by nationality and by level of study, i.e. taught master’s (dark and light purple blocks), research master’s (green block), first degree (undergraduate), and other categories. This data provides us with two insights. First, we translate these stock data to impute entry flow data using typical program length, particularly for research postgraduate and undergraduate students. Then we net these flows out, together with the small below-degree and no-show blocks introduced above, to obtain total entry into taught master’s programs and the college sector, given by the sum of the purple, orange, and brown blocks in Figure 27.

Second, after aligning HESA (academic years) with visa cohorts (calendar years), we subtract the visa-based measure of HEI entry, constructed from the previous paragraph, from the HESA-based taught-master entry. The residual (the light-purple segment) measures students entering through franchised colleges. Franchised colleges stand for private partner institutions that recruit and teach students toward a university-validated master’s degree. Because these students are formally registered for the university’s award, HESA includes them in the university’s taught-master numbers; but because partner colleges issue their CAS, visa statistics record them under “other institutions” rather than HEIs. The channel thus appears in HESA enrolment data while being invisible in HEI visa data in CAS. In

³⁰We do not directly use the information on visa applications to other college sectors in this dataset, as the grant rate for this type of application is unclear. Instead, we use them as a cross-check: the implied grant rate for college applications rises from about 67% in 2011 to over 90% by 2013, consistent with the post-2012 college sector consisting of the compliant survivors of the reforms.

practice it operated much like the low-skilled bogus-college route, off-campus recruitment piggy-backing on universities’ degree-awarding powers. The 2011–12 reforms, which required all sponsors to obtain Highly Trusted Sponsor status and independent educational oversight, closed it: our estimate of the franchise segment falls from roughly 10,000 (2010 cohort) to essentially zero by 2013. Furthermore, since we have identified the sum of franchised college and other college sectors through the first and second datasets, we also separately identify the numbers in college sector (brown) and franchised college (light purple).

We supplement these primary sources with several auxiliary Home Office series. Visa outcome statistics provide approval rates for Indian study-visa applications, which we use to convert sponsorship applications (source 2 above) into actual entries. Visa records disaggregated by course level (only exist post-2018) allow us to strip undergraduate and research-degree students out of aggregate study-visa counts at the quarter level. The same records also indicate the share of recent Indian study visas that are granted for courses below degree level (the teal block). For the grey block of “no-shows”, the gap between total visas issued and the *Migrant Journey* entry cohort, about 3 to 4 percent per year, bounds the share of granted visas never used for entry. In-country extension statistics showing granted extensions by nationality complement the application counts in the second source mentioned above. They pin down the application-to-grant conversion rate of extension visas and extend the extension series to the years before CAS begins in 2010. Finally, because the CAS sponsorship system begins in 2010, for earlier cohorts we allocate total study-visa issuance between the two sectors using an imputed HEI share. Details of construction are available upon request.

It is worth being explicit about what these data do and do not identify. Some objects are observed directly: for each entry cohort, the *Migrant Journey* data record end-of-year visa status: how many students remain in the UK on study leave, how many transition to work visas, and how many leave. Enrollment data provide counts of taught-master students (inclusive of those registered through franchised colleges), research postgraduates, and undergraduates. A second set of objects is identified only under data-driven assumptions. For example, combining sponsorship applications with calibrated approval rates lets us decompose the taught-master total into students entering HEI programmes directly and those entering through franchised colleges. A final set of objects the data cannot deliver at all: the joint distribution of entry channel and subsequent visa path. Visa-status transitions are observed only at the cohort level and institution type only at entry; no source links the two at the individual level. Thus, we cannot estimate, for example, the share of franchise students who ultimately stayed on work visas relative to their university-track peers.

I.2 Results

We use 2011 as an illustrative example and flow charts for other years are similar. A total of 28,281 Indian students first entered on a student visa in calendar year 2011: approximately 537 research postgraduates, 2,208 undergraduates, 5,184 in the other college sector, 1,024 students enrolled in below-degree programs in HEI, and 339 students did not show up. Among the remaining 16,683 students in taught master's programs at HEI, nearly half of them (8,037 of 16,683) were enrolled in franchised taught master's programs. By the end of 2012, 6,557 students had ended their student visa, and 1,344 had obtained a UK work visa, most of them Tier 1 PSW visas (82%, 1,096 of 1,120), drawn from the spring 2011 entry cohort on taught master's programs or in the other college sector. By the end of 2013, all taught master's students should have graduated, but only 461 obtained a work visa in that year, of which only 48 were Tier 1 visas (10%), reflecting the end of the Tier 1 PSW route. At the end of year 2, approximately two thirds (11,102 of 17,560) left the United Kingdom, while approximately one third (5,789 of 17,560) remained on a student visa, most plausibly enrolling in a second master's program.

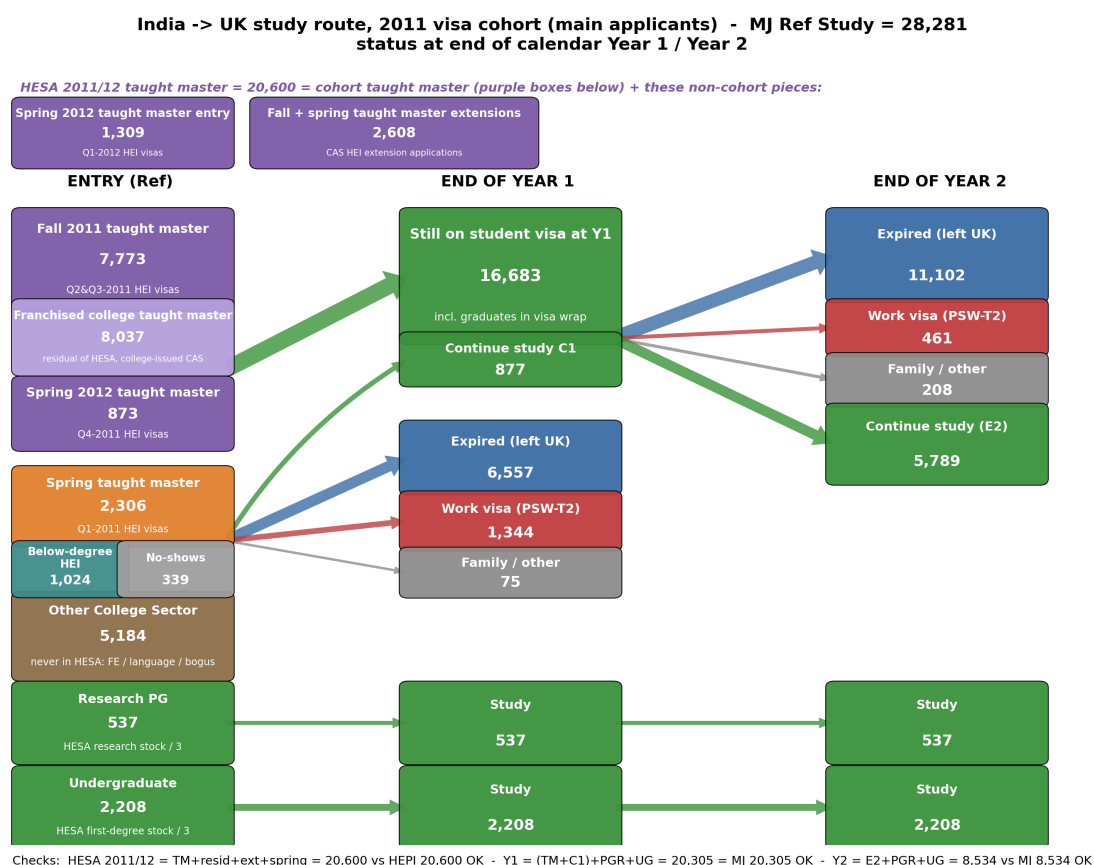


Figure 27: UK Study Route (2011)

I.2.1 Insight on Education

Figure 28 plots entry into taught master’s programs at HEIs (the dark-purple and orange blocks) against entry into the college sector and franchised provision (the brown and light-purple blocks), where low-quality and fraudulent provision was concentrated. Enrollment in the low-end education collapses from 23,259 in 2010 to only 347 in 2013, a 98.5 percent decline driven by both sides of the market. In the demand side, the PSW closure removed the employment payoff these students sought; in the supply side, license revocations to education institutions under the Highly Trusted Sponsor regime shut the providers themselves. Entry into genuine taught master’s programs also falls, but far less steeply, from 14,272 to 7,033, or 51 percent, and by almost exactly the same proportion as Indian UK master’s counts in our LinkedIn sample (See LinkedIn coverage across years in Appendix I.3), reassuring us that LinkedIn coverage of this population is stable across the reform window. The reform thus did most of its work on where it was aimed, with the “college sector and franchised provision” segment falling from 62 percent of Indian entries in 2010 to under 5 percent by 2013. However, it carried a sizable unintended cost that high-skilled enrollment halved as the high-skilled employment opportunities also shrank associated with the removal of the unconditional stay visa.

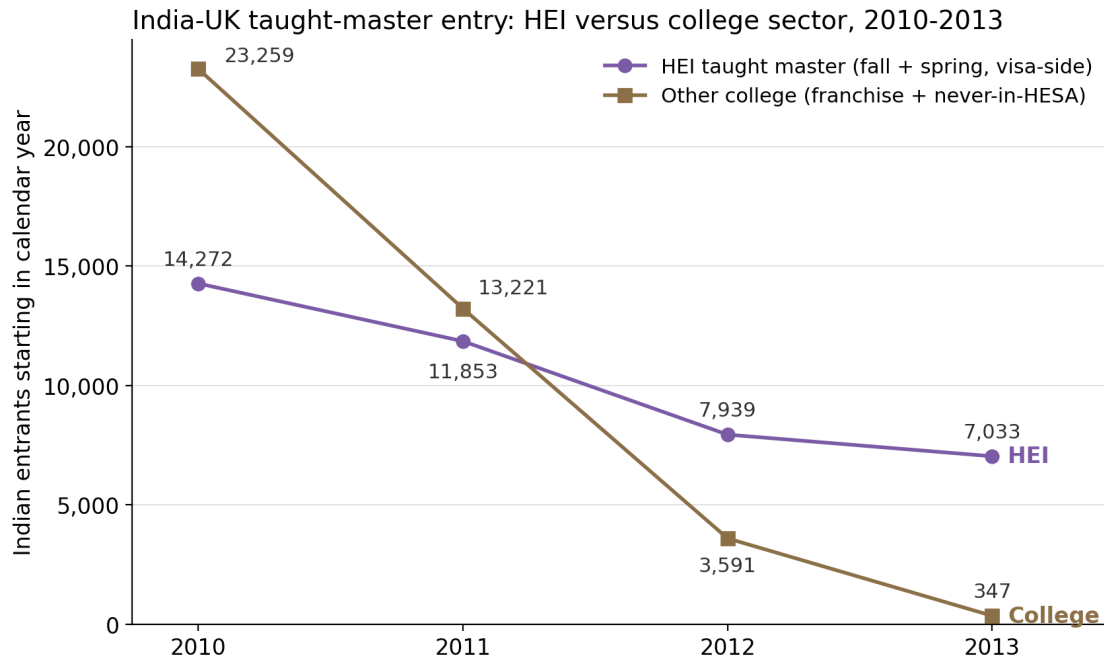


Figure 28: Trend of Enrolled Indian Students by Visa Sponsor Type

I.2.2 Insight on stay in UK for work

In Figure 29, the blue line plots, for each cohort, the total number of Indian students who stay in the UK after their studies (transitions into work visas plus family and other leave, summed over years one and two) and the orange line plots the subset transitioning into Tier 2, the employer-sponsored high-skilled work visa. Several important takeaways emerge from this figure. First, the blue line collapses from 11,692 for the 2010 cohort to 724 for the 2012 cohort, reflecting the removal of the PSW visa. Before the reform, staying on was overwhelmingly PSW-based. The 2011 cohort, which straddles the transition (Figure 27), reveals the mechanics. They hold 1,344 work visas at the end of year one (against 3,222 for the 2010 cohort), but add only 461 in year two (against 8,151). This shows that some spring 2011 entrants completed their one-year programs in time and managed to claim PSW before the April 2012 closure, while the fall majority, graduating in late 2012, faced a world in which employer sponsorship Tier 2 visa was the only remaining route. In fact, Tier 2 visa covers most of the blue line post-reform: 490 of the 724 stayers in the 2012 cohort and 635 of the 794 in 2013. Moreover, family and other non-work routes play essentially no role in these movements. They account for only 100 to 320 stays per cohort throughout. Finally, the stability of the orange line does not imply that the flow of high-skilled stayers was stable. Even for high-skilled graduates, most of them exploited PSW rather than Tier 2 visa to stay before the reform. Thus, Tier 2 counts alone cannot reveal how many high-skilled workers the UK lost. This is precisely why we turn to LinkedIn data to pin down the reform’s impact on high-skilled labor.

I.3 Insight on for LinkedIn coverage of high-skilled workers

Figure 30 reports the LinkedIn coverage rate for our population of interest, high-skilled Indian students. The purple line plots the number of Indian taught master’s students in the U.K. we observe on LinkedIn as a share of the total genuine Indian taught master’s intake reported in the purple line of Figure 28. The orange line in Figure 30 plots the number of Indian master’s graduates observed on LinkedIn as stayers, as a share of the total number remaining in the United Kingdom reported in Figure 29. From this figure, we see that the purple line is flat at approximately 30%, indicating that the reform did not shift the representativeness of the LinkedIn sample among enrolled students. On the contrary, the orange line rises sharply across the reform due to the effect of the policy. Before the reform, a substantial share of stayers worked in low-skilled occupations and did not use LinkedIn. After it, stayers were predominantly high-skilled, and thus the LinkedIn coverage rate rose accordingly.

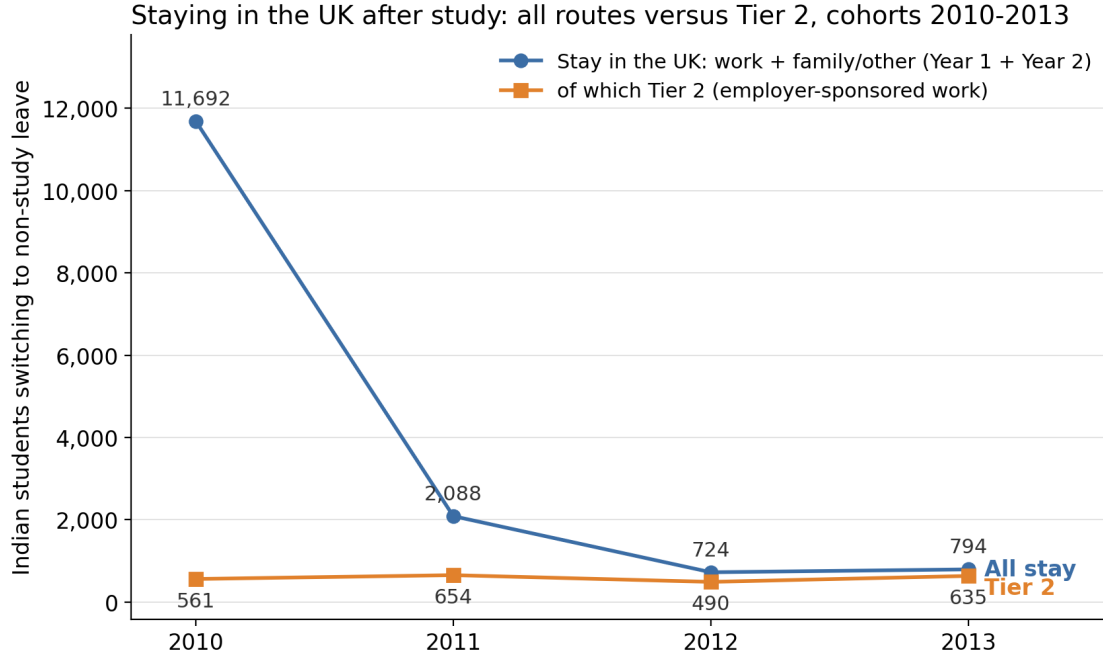


Figure 29: Trend of Stay

Assuming the majority of Indian stayers post-reform operate in the high-skilled sector, their LinkedIn coverage rate stands at slightly above 40%. This number is roughly comparable with Khanna and Morales (2025), who benchmark LinkedIn data against the American Community Survey (ACS) and find a 54.3% coverage rate among college-educated Indian immigrants in the U.S. However, a critical identification challenge arises from differential platform penetration across destinations, as captured by the gap between the orange and purple lines. Specifically, students who stay in the U.K. have a systematically higher LinkedIn coverage rate than students leaving the U.K., who are most likely returning to India. (Amanzadeh et al., 2024) also mentions the gap in LinkedIn penetration between developed economies and India, suggesting a coverage rate of roughly 40% for India’s skilled workforce, compared to over 80% in many developed countries. This missing-not-at-random (MNAR) discrepancy in the stage-two location choice introduces a selection bias into our model estimates. To correct for this, we may adopt the “inverse weighting” approach exploited in Amanzadeh et al. (2024), upweighting returning students to offset the platform’s lower coverage rate in their destination.

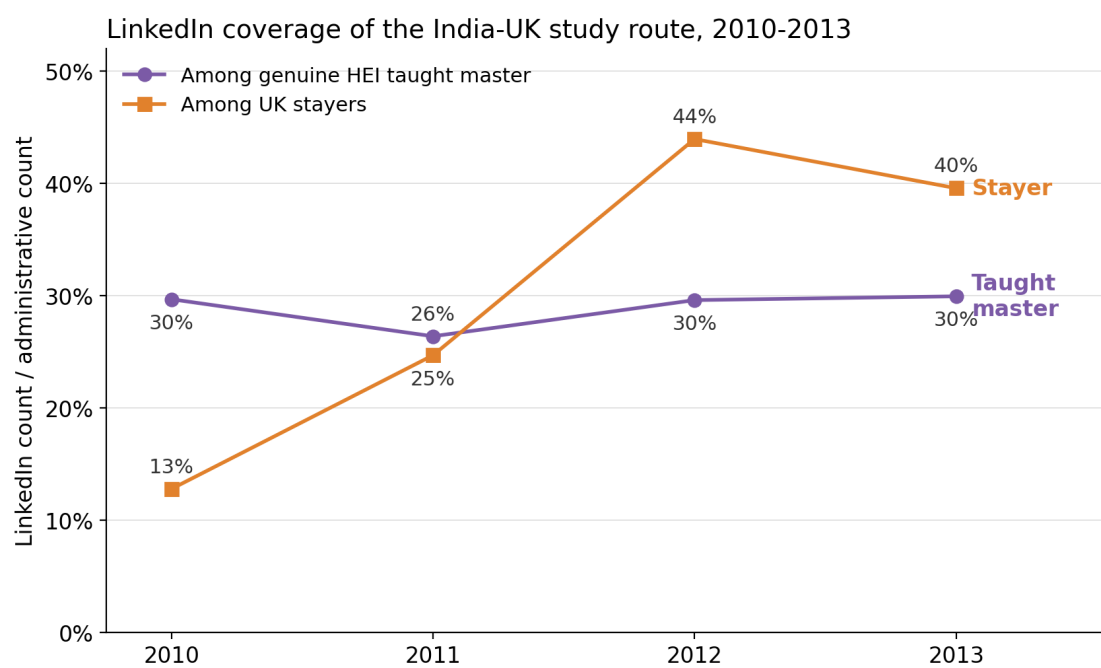


Figure 30: LinkedIn Coverage Rate